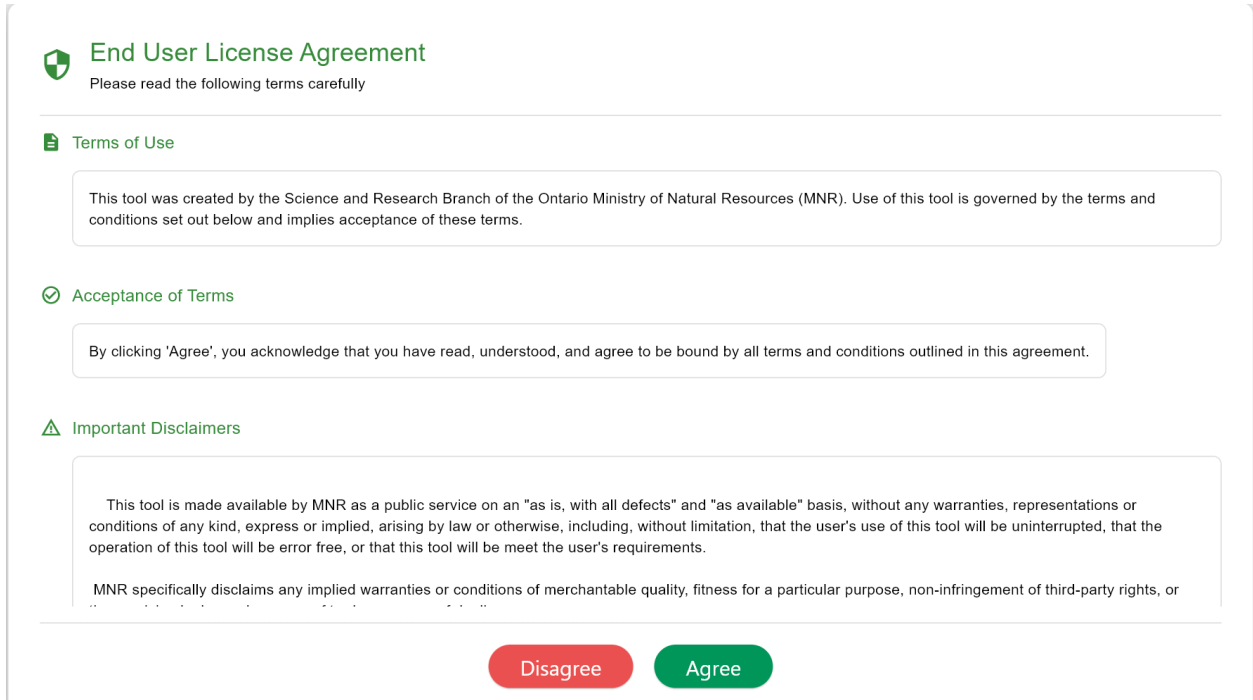


Index

1. [EULA \(Pg. 2\)](#)
2. [About and Supported Species \(2\)](#)
3. [Select Data \(7\)](#)
 - a. [Importing from Text File \(7\)](#)
 - b. [Importing from Database \(9\)](#)
4. [Calculate Biomass \(15\)](#)
 - a. [Selecting Equation + Components \(16\)](#)
 - b. [Species Classification \(19\)](#)
 - c. [Exporting Results \(21\)](#)
 - i. [Exporting to Text File \(22\)](#)
 - ii. [Writing to Database \(24\)](#)
 - iii. [Generating Plot \(25\)](#)
 - d. [Missing Species? \(27\)](#)
5. [Create Species \(27\)](#)
6. [Modify Species \(31\)](#)
 - a. [Search for Species \(32\)](#)
 - b. [Species Details \(32\)](#)
 - c. [Edit Species \(34\)](#)
 - d. [Delete Species \(36\)](#)
7. [Settings \(38\)](#)
8. [Exit Application \(39\)](#)
9. [Log \(42\)](#)

1) EULA

To use the program you must agree to the End User License Agreement (EULA). Click “Agree” to continue to the Main Menu. Click “Disagree” will close the app immediately.



End User License Agreement
Please read the following terms carefully

Terms of Use

This tool was created by the Science and Research Branch of the Ontario Ministry of Natural Resources (MNR). Use of this tool is governed by the terms and conditions set out below and implies acceptance of these terms.

Acceptance of Terms

By clicking 'Agree', you acknowledge that you have read, understood, and agree to be bound by all terms and conditions outlined in this agreement.

Important Disclaimers

This tool is made available by MNR as a public service on an "as is, with all defects" and "as available" basis, without any warranties, representations or conditions of any kind, express or implied, arising by law or otherwise, including, without limitation, that the user's use of this tool will be uninterrupted, that the operation of this tool will be error free, or that this tool will be meet the user's requirements.

MNR specifically disclaims any implied warranties or conditions of merchantable quality, fitness for a particular purpose, non-infringement of third-party rights, or

Disagree **Agree**

Figure 1. EULA screen

2) About and Supported Species

There are two highlighted buttons: [About] and [Supported Species]. On the left-side column, clicking on the purple-highlighted button [About] to see the general information of the tool. Once the pop-up shows, click on [X] button on top left corner to go back to Main Menu.

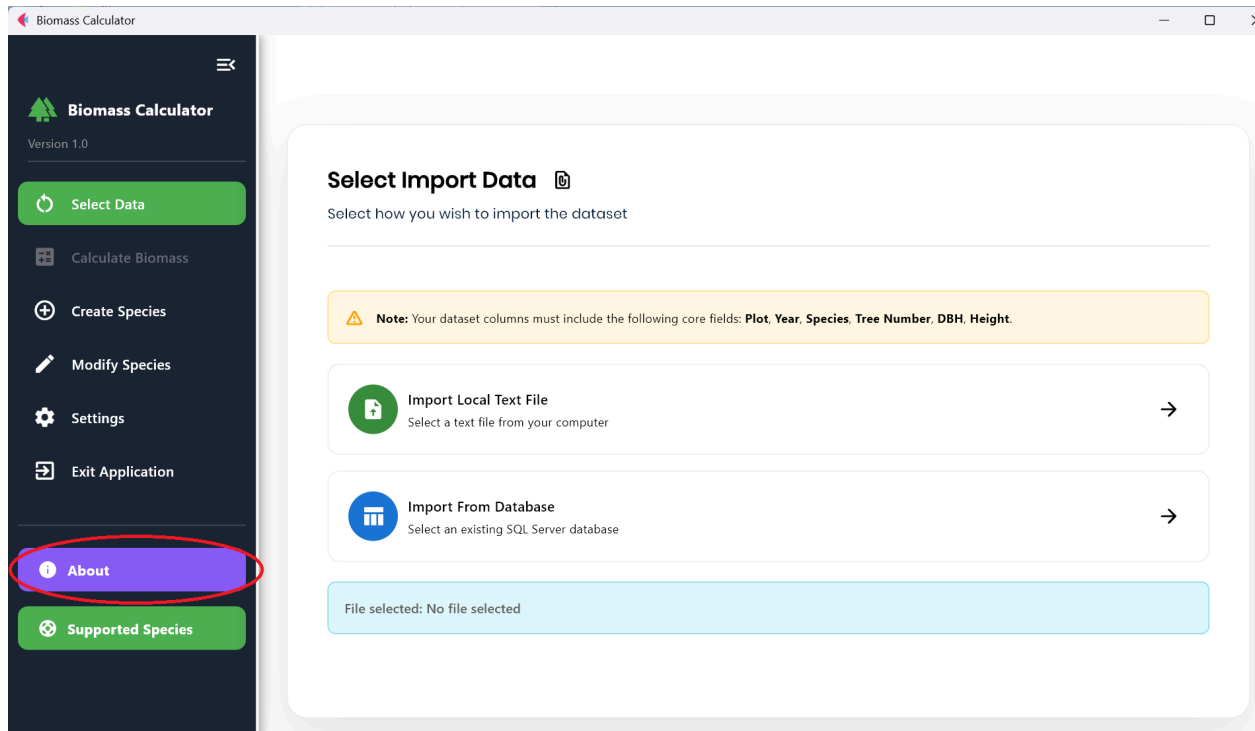


Figure 2. [About] button highlighted.

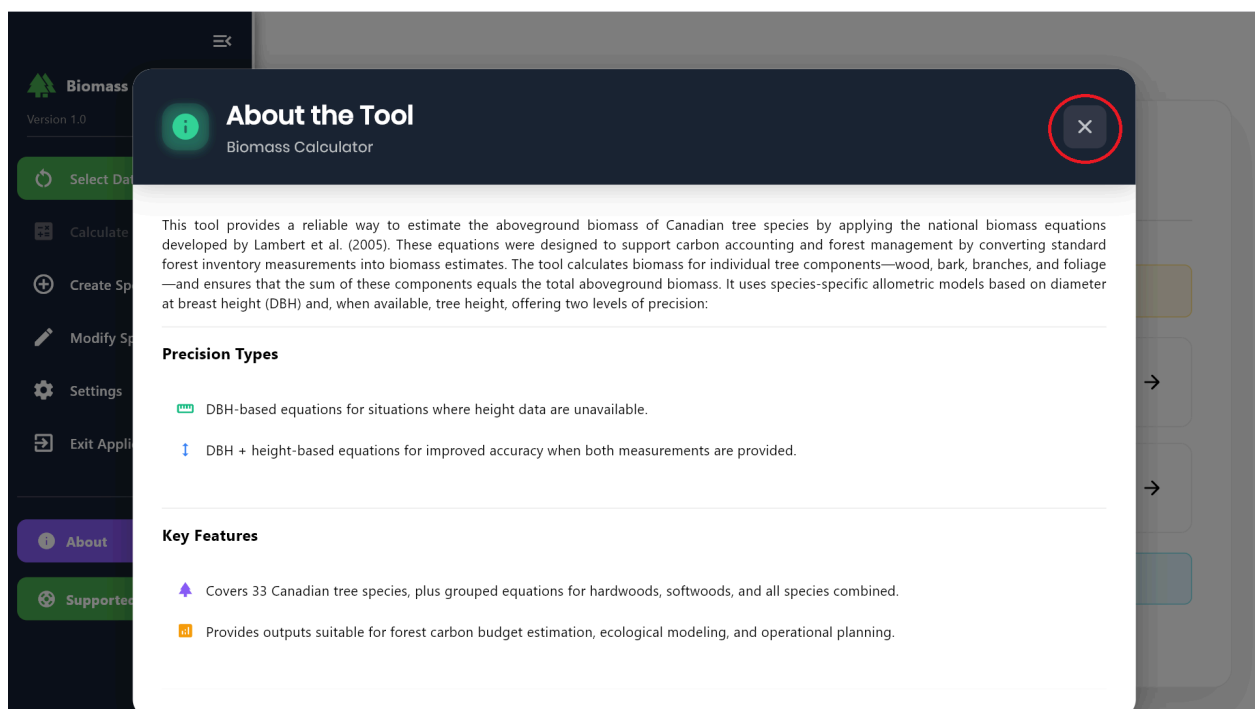


Figure 3. [About] section pop-up. [X] button highlighted.

The green-highlighted button [Supported Species] shows a look-up section that users can find all tree species and associated code numbers that are currently supported by this tool for biomass calculations. Once the pop-up section shows up, click on [Dismiss] at the lower right corner of the section to go back to Main Menu.

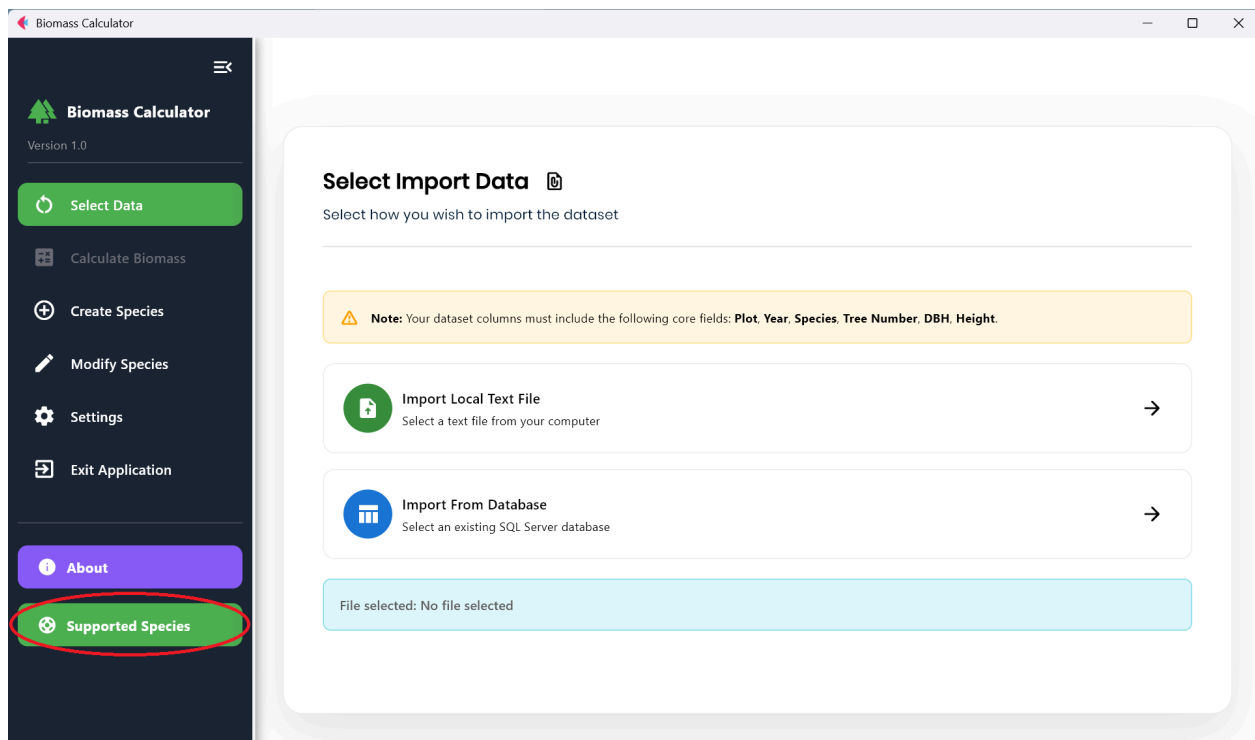


Figure 4. [Supported Species] button highlighted.

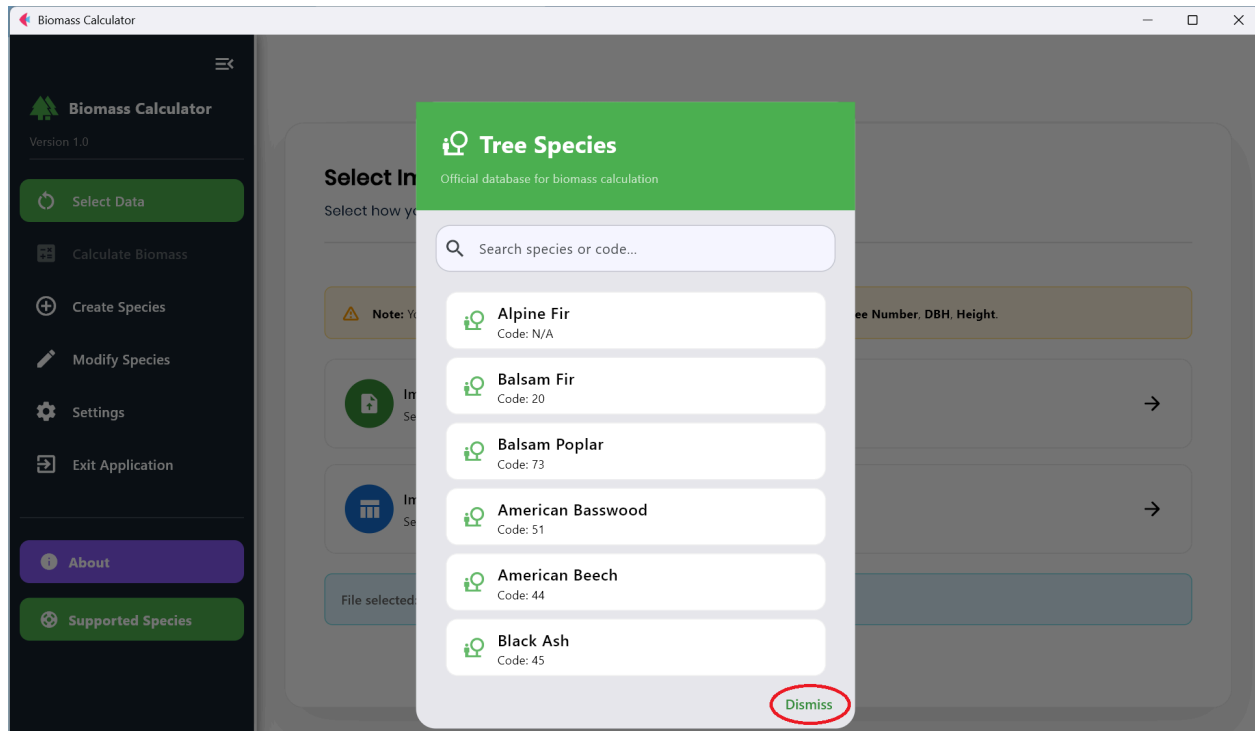


Figure 5. [Supported Species] section pop-up. [Dismiss] button highlighted

Within [Supported Species], users can search species by name or code number. If the user types alphabet character(s), the function will find all matching species names that contain such string of characters (case insensitive).

On the other hand, if the user types a number, the function will try to find the species that either has the matching code number or contains that number in its code number.

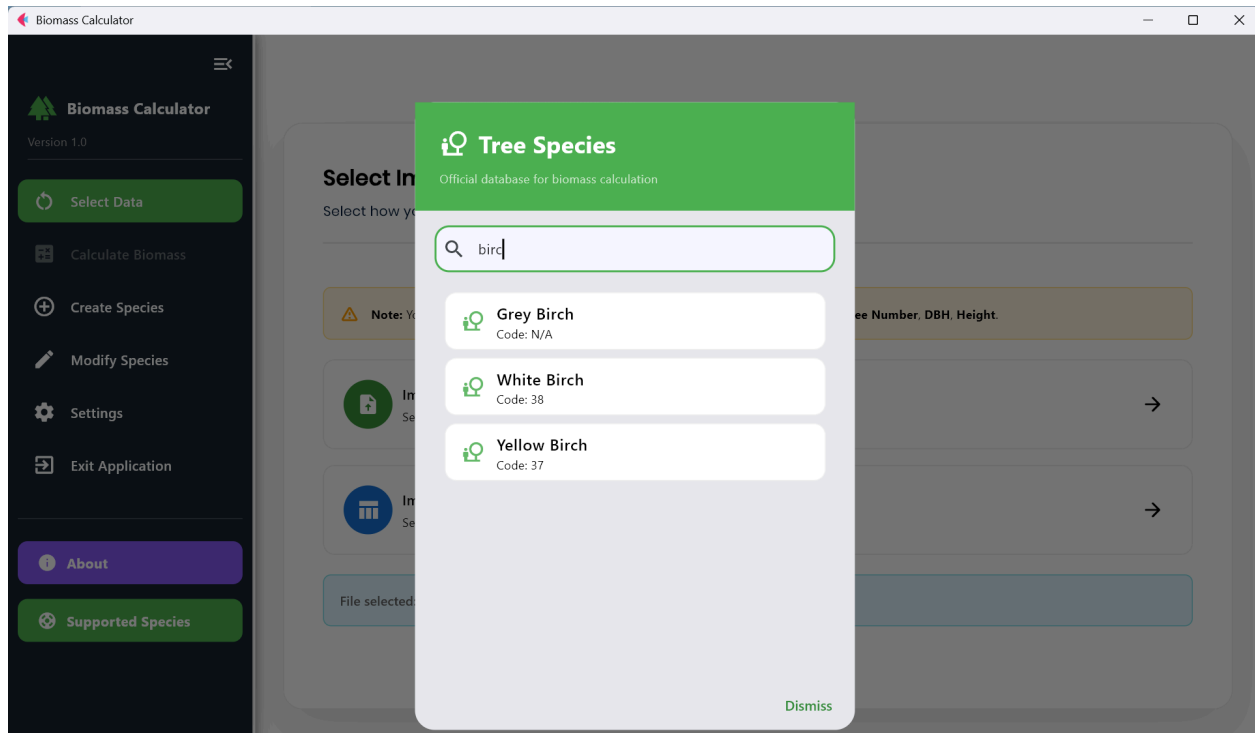


Figure 6. [Supported Species] section pop-up. Species look-up by name string.

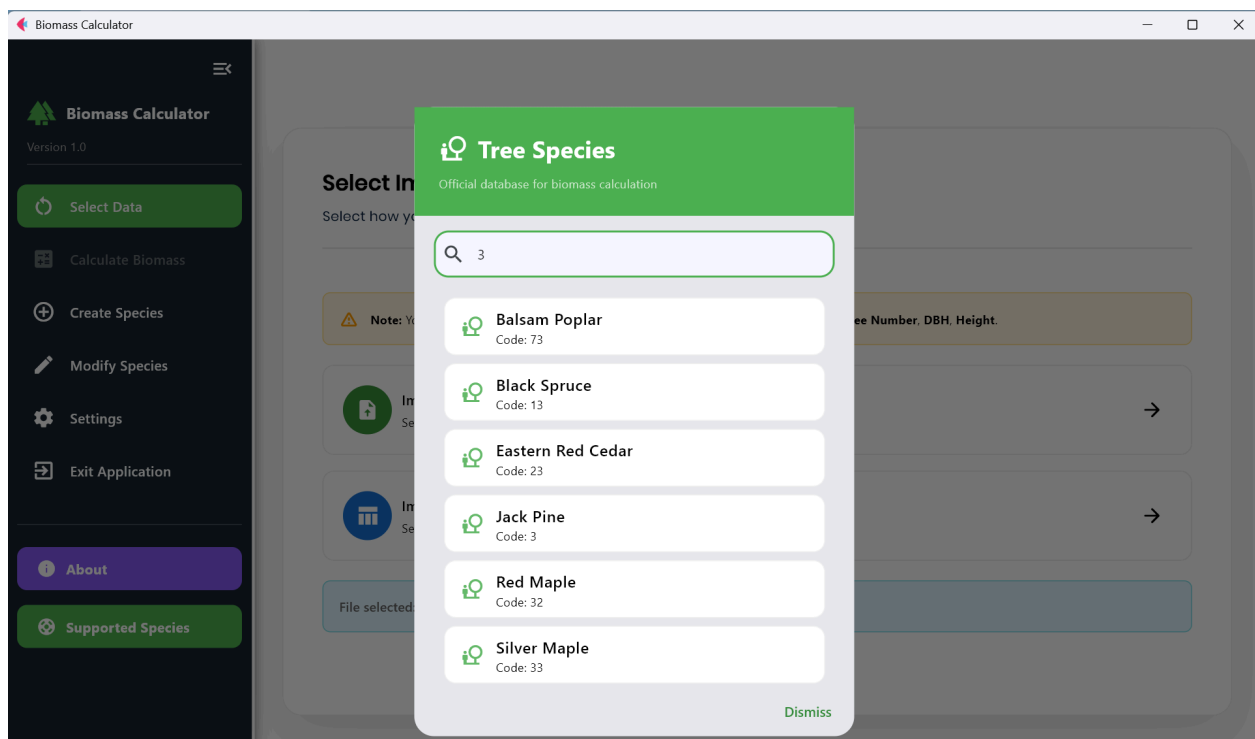


Figure 7. [Supported Species] section pop-up. Species look-up by code number.

3) Select Data

To select the data to be used, open the “Select Data” menu by clicking on the Select Data button on the left sidebar.

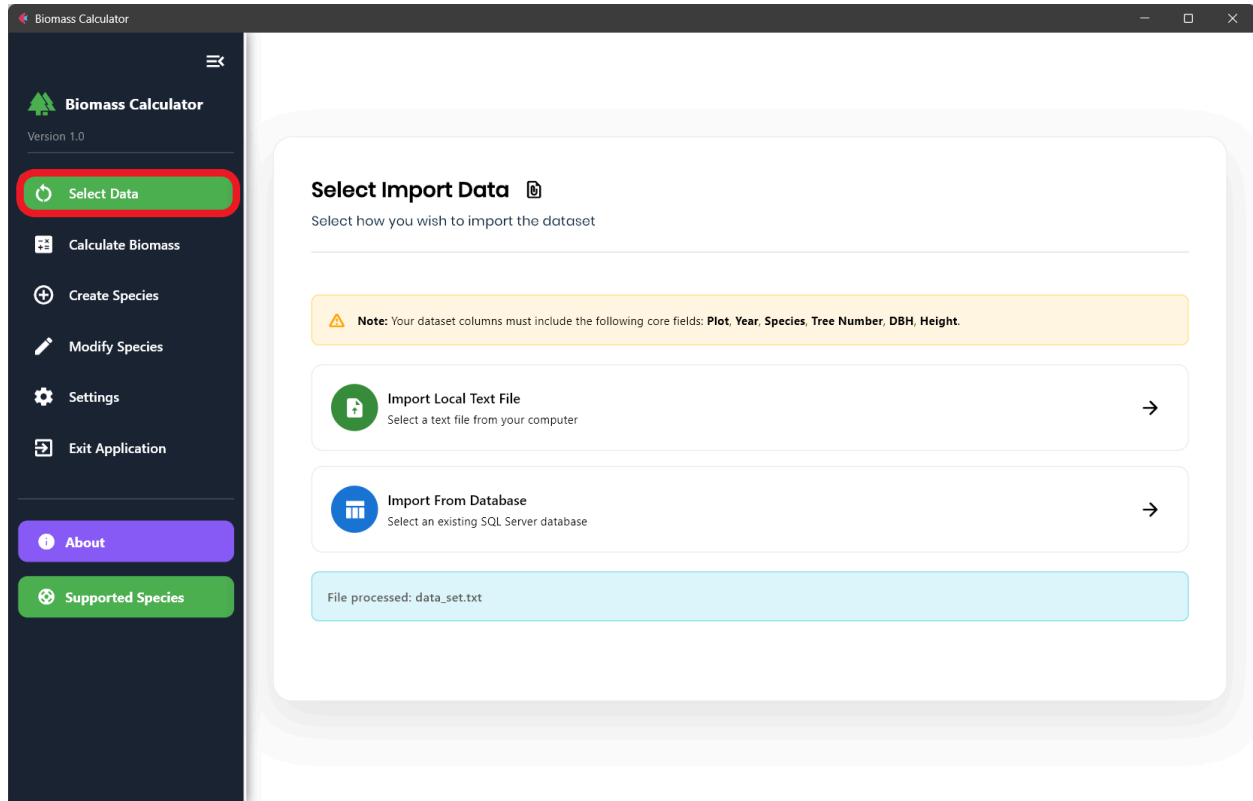


Figure 8. The [Select Data] button is highlighted.

a) Importing from Text File

To import a text file, select [Import Local Text File] which will prompt you to select a text file to use.

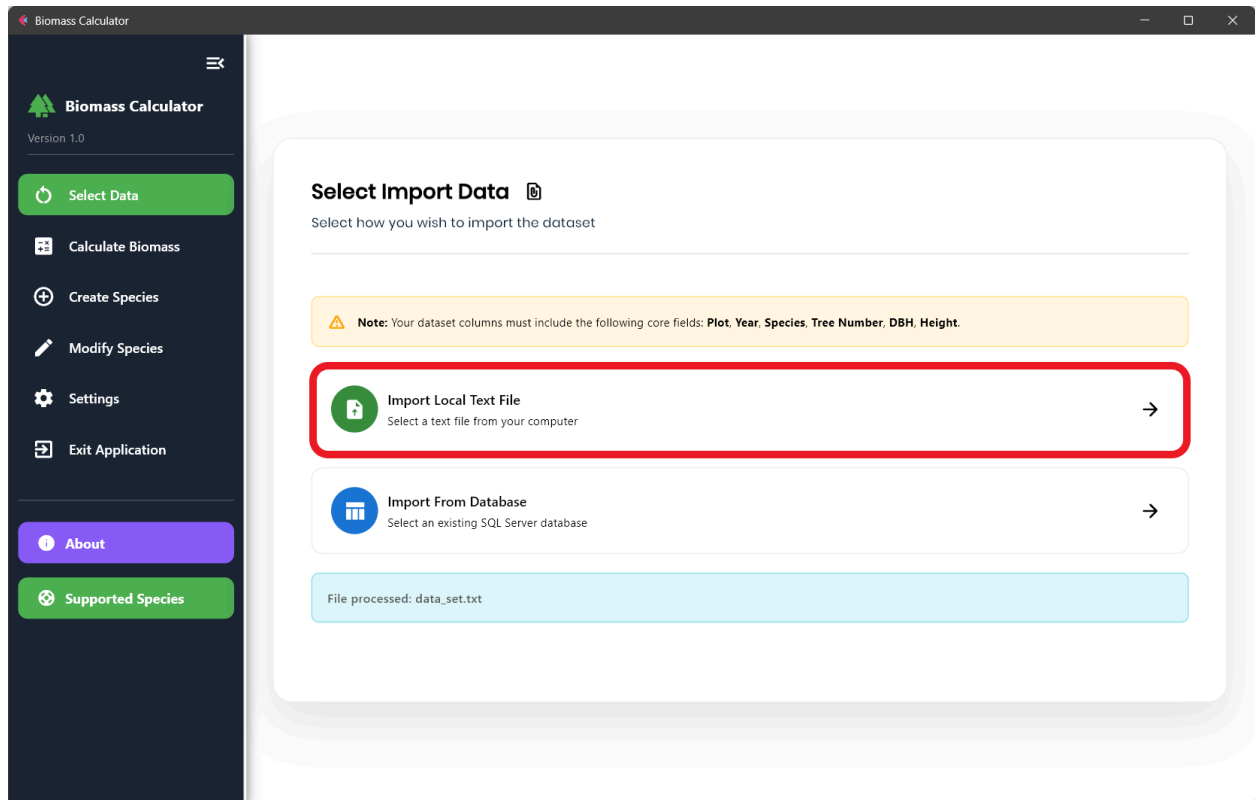


Figure 9. The [Import Local Text File] button is highlighted.

The dataset columns in the text file must have the core fields:
Plot, Year, Species, Tree Number, DBH, Height.

For example:

Plot	SubPlot	Year	Origin	Tree Status	Species	Tree Number	DBH	Height
1108656	1	2005	Natural	Stand L	251	2	18.3	14.48

Figure 10. Screenshot of the fields in a text file.

If done correctly it should show the name of the file selected at the bottom.

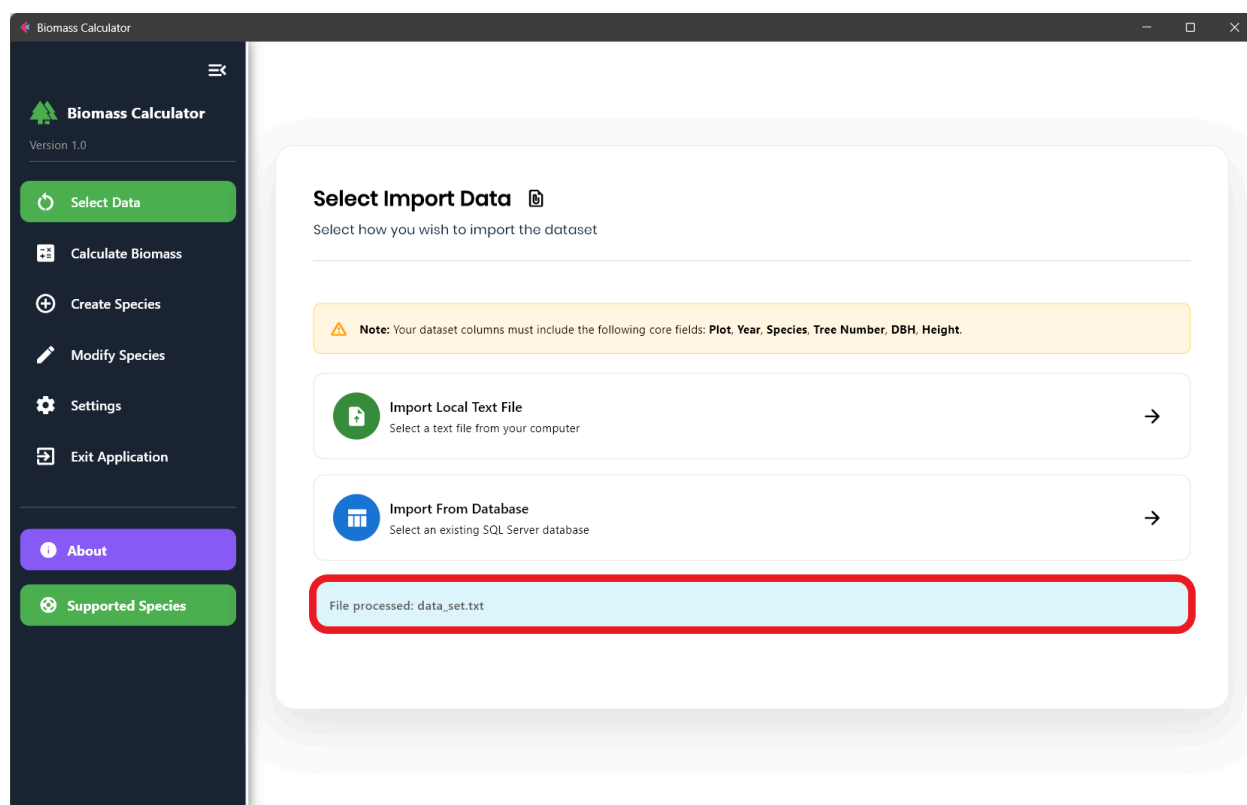


Figure 11. The confirmation of the file processed is highlighted.

b) Importing from Database

To import from a database select “Import From Database” which will open a menu to enter details to connect to a database. In order to select this option, users’ machines need to host a SQL server, so the tool can connect to the database directly. Preferably, users should have Microsoft SQL Express installed as SQL server engine, and Microsoft SQL Server Management Studio as the Graphical User Interface to interact with SQL server.

Once host machines have both software installed and executed, the tool will automatically detect the SQL server engine. Users just need to type in the name of the database needed for biomass calculation, and the tool will open connection to that database to get input data from and write output data to.

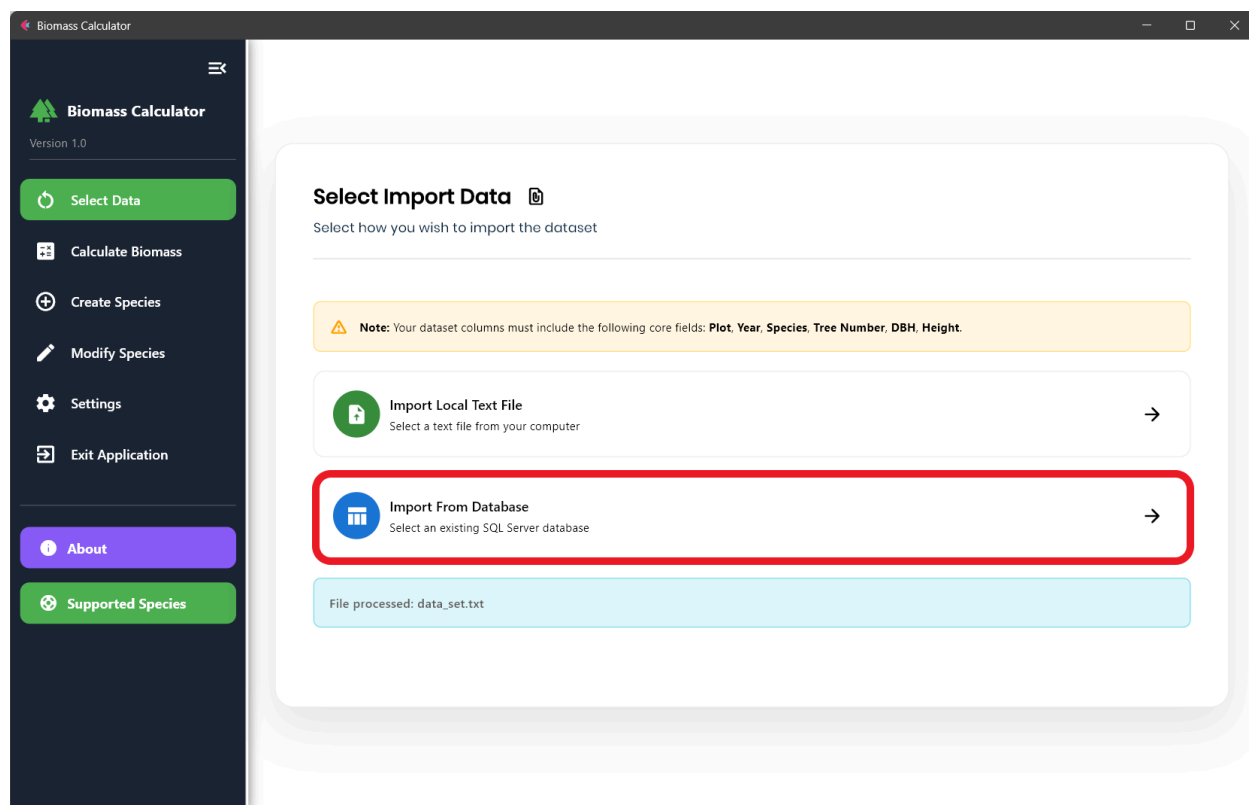


Figure 12. The [Import From Database] button is highlighted.

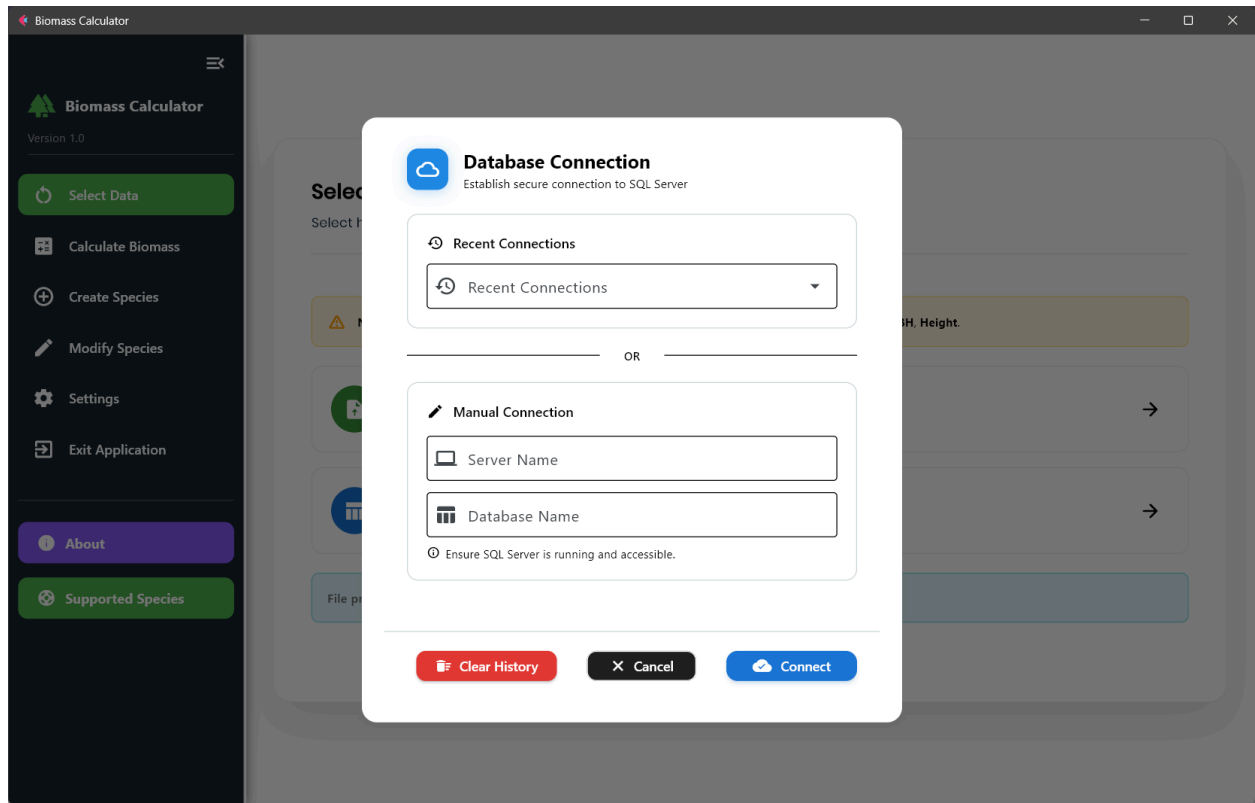


Figure 13. The opened menu for selecting a database (no SQL server detected so all fields are empty)

To select a database, you can either select a recently connected database by opening the drop-down menu and clicking on one of the connections, or you can manually enter a server name and database name into the corresponding text boxes to connect to (in case the tool cannot detect any running SQL server).

****Note**:** In case users connect to “gyPSPPGP” database, the input data comes from “dbo.tCalcBiomassInput” table. The table for storing calculated biomass data is named “dbo.tCalcBiomassOutput”

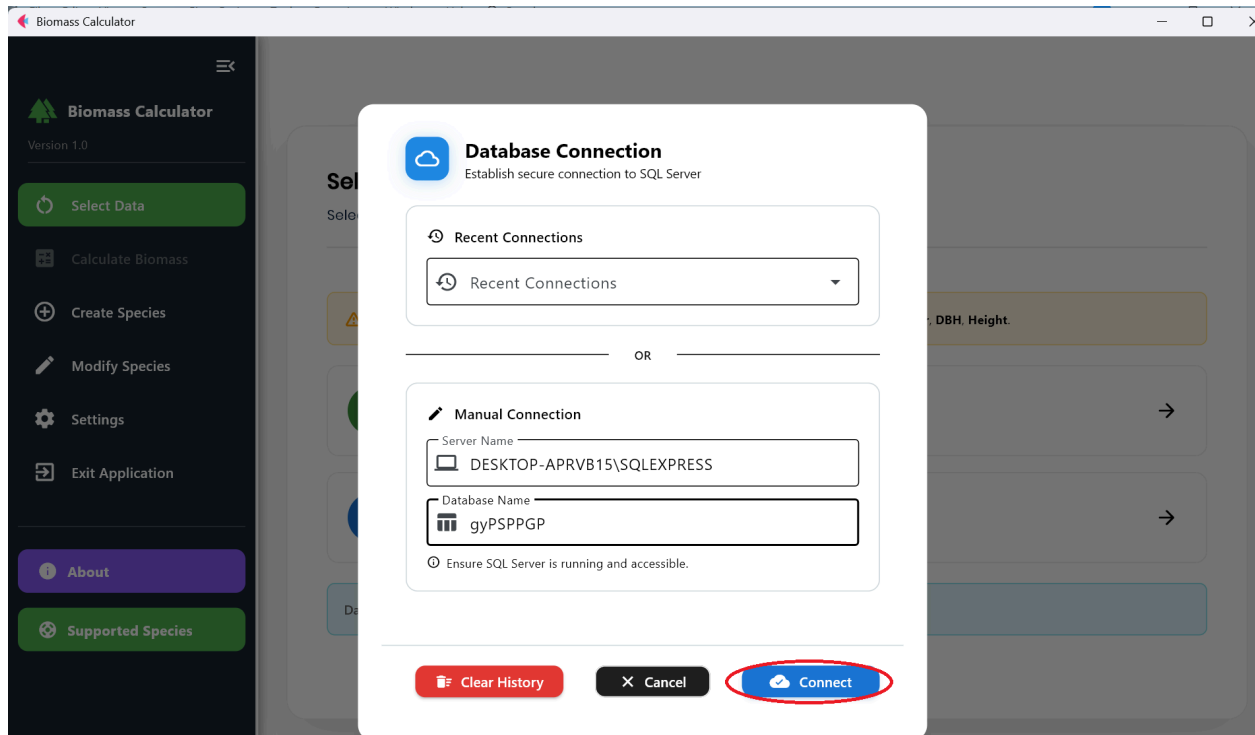


Figure 14. Manually enter “Server Name” and “Database Name”, and then click [Connect] to open connection for the first time (no connection history)

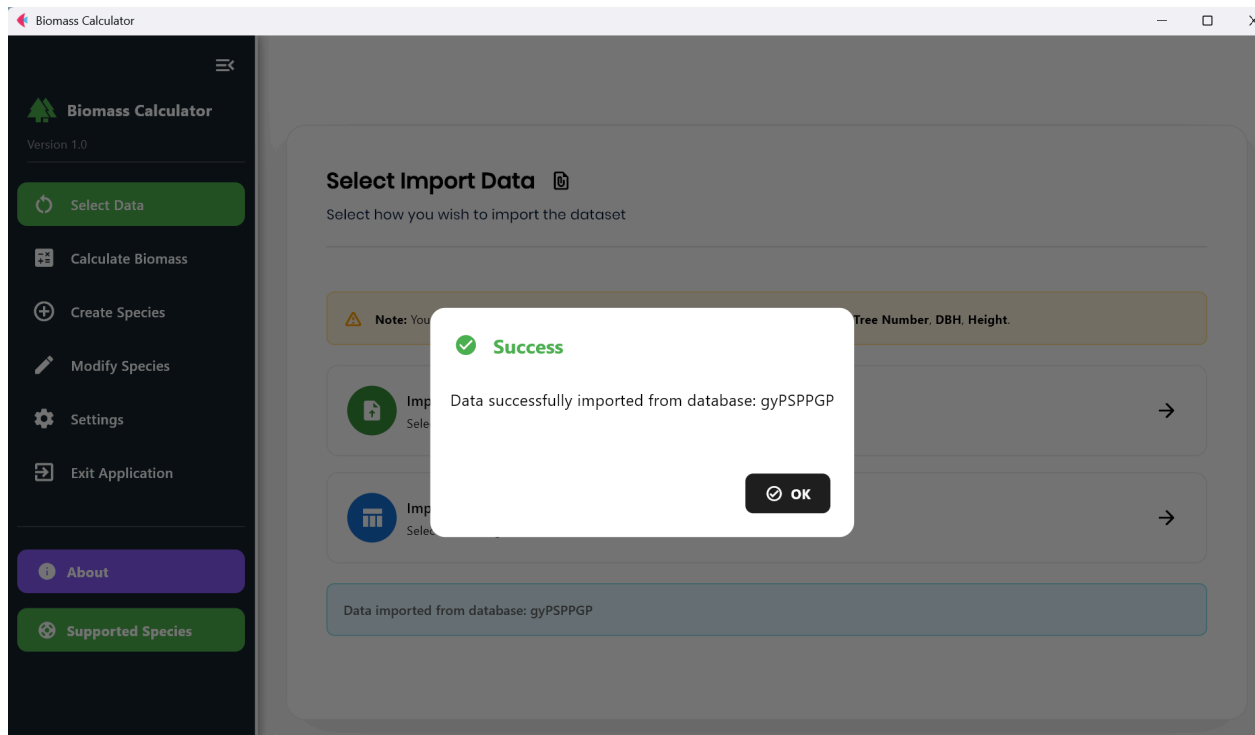


Figure 15. Notice when the tool successfully connects to a database and imports data from it

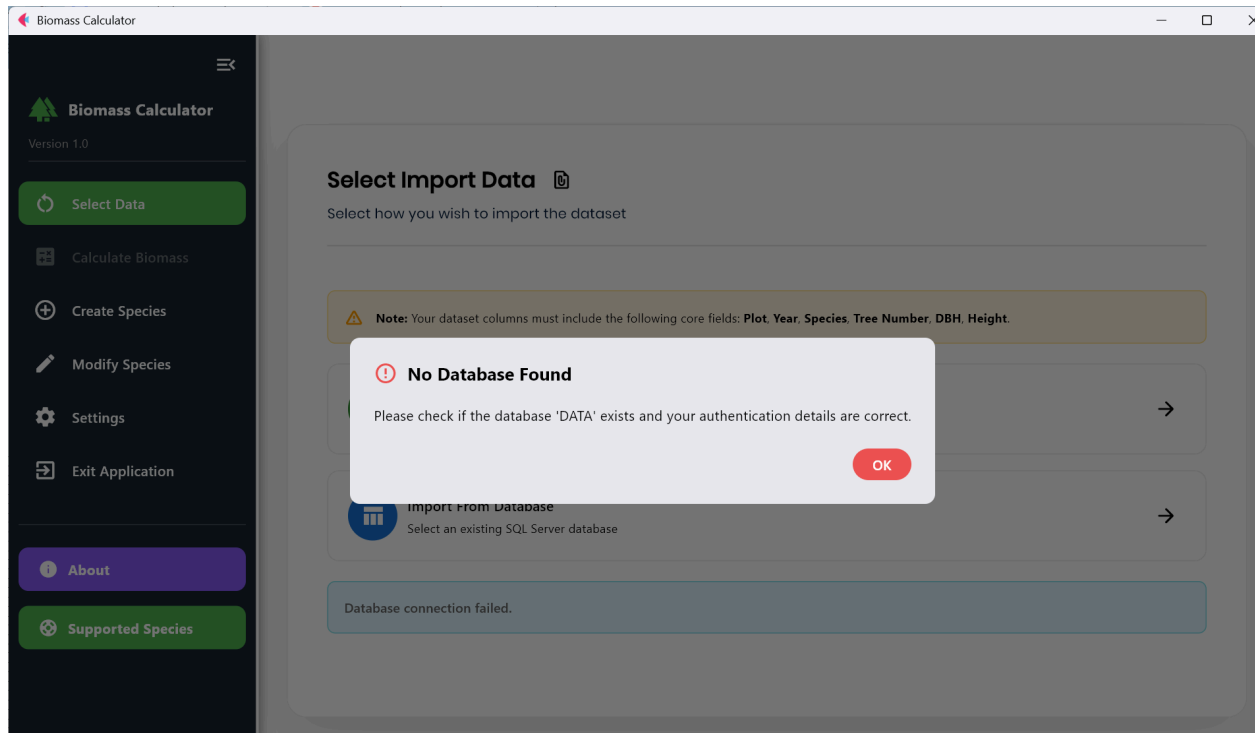


Figure 16. Notice when user enters incorrect server name and/or database name, or doesn't have permission to access such database

If the tool made connections in the past, clicking on the drop-down arrow in "Recent Connections" will show all connection logs. Clicking on any will auto-fill the fields "Server Name" and "Database Name", then click on the [Connect] button to confirm.

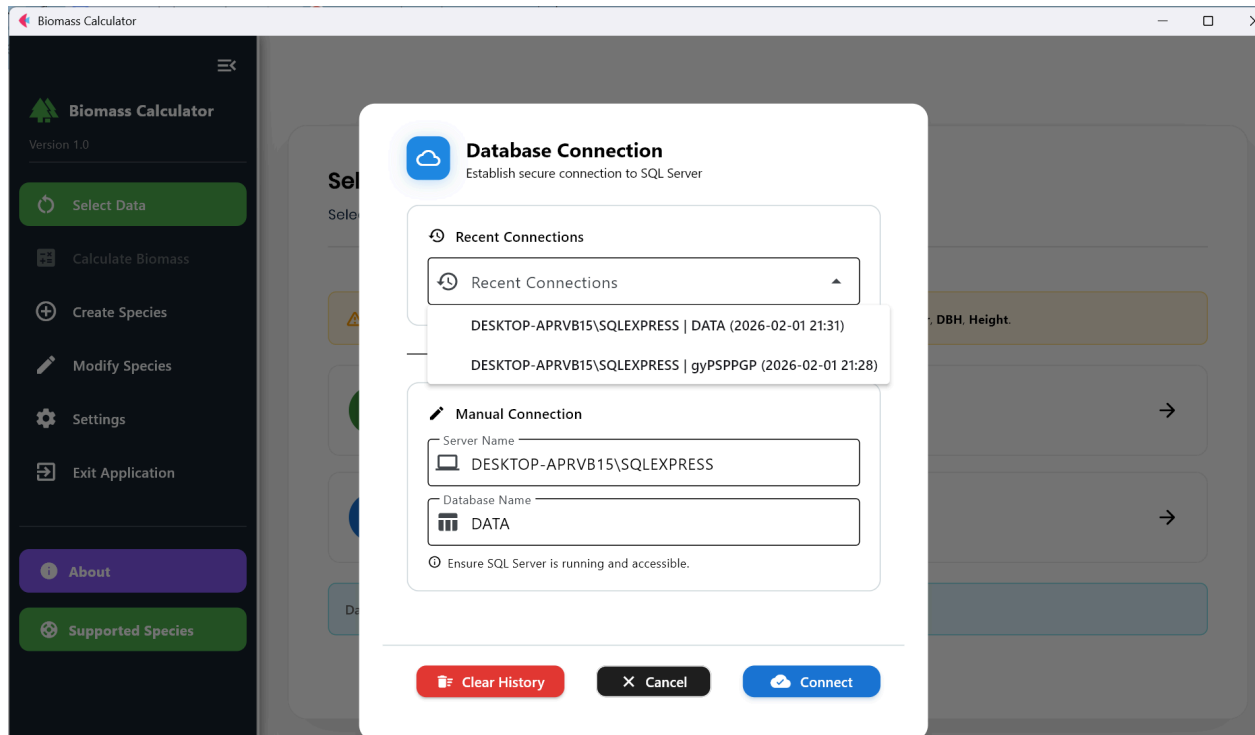


Figure 17. “Recent Connections” shows logs

To clear the history of all recent connections, click on the red “Clear History” button. After performing that, clicking on the drop-down arrow will show no logs.

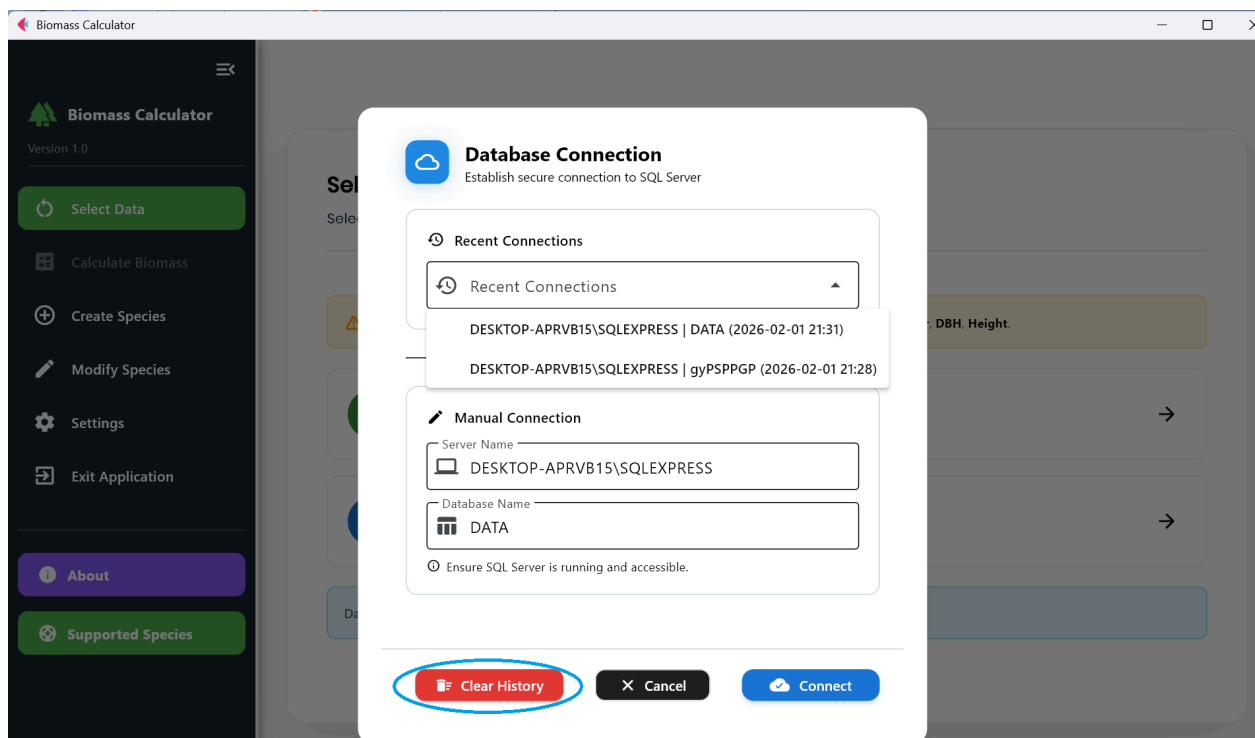


Figure 18. Before clearing “Recent Connections”. The [Clear History] button is highlighted.

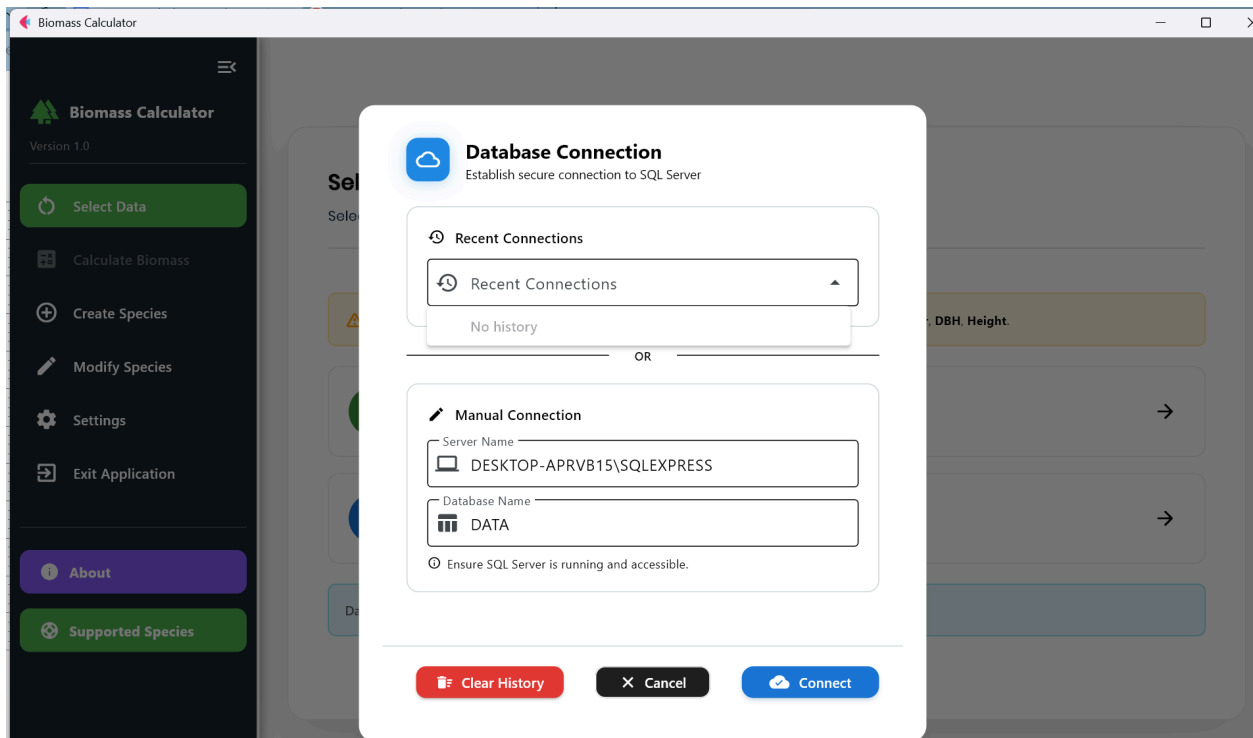


Figure 19. After emptying connection history.

4) Calculate Biomass

After selecting input data, click on [Calculate Biomass] button on the left-hand side menu specify which type of computation to perform on the data

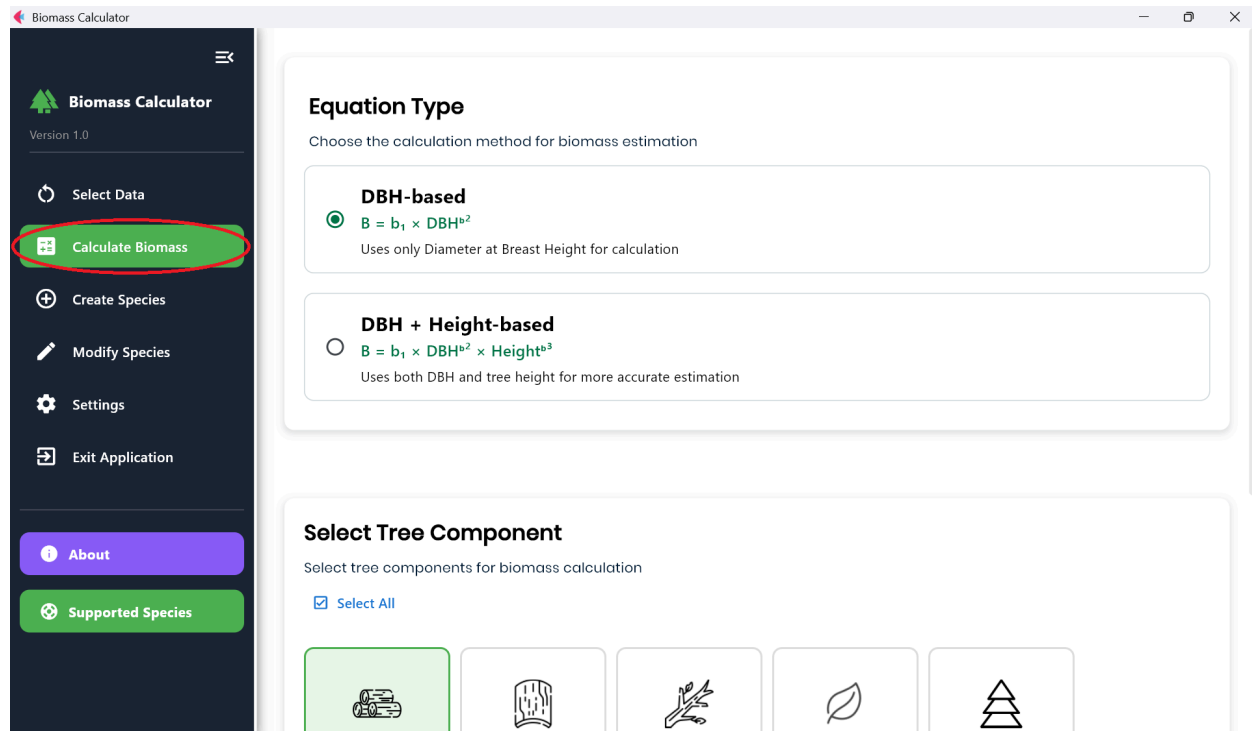


Figure 20. App screen after clicking on [Calculate Biomass]

a) Selecting Equation + Components

In the “Equation Type” section of the [Calculate Biomass] screen, users can choose one of the two biomass-calculating functions simply by clicking on the one desired: DBH-based (simple) or DBH-and-Height-based (more accurate). Default selection is DBH-based

Also, users can select to have all biomass features to be accounted for in the equation, or select some features and omit the rest for calculation. A feature when selected (by clicking on it) has the green background and outline, and clicking on that feature again deselects it (and its background turns to white)

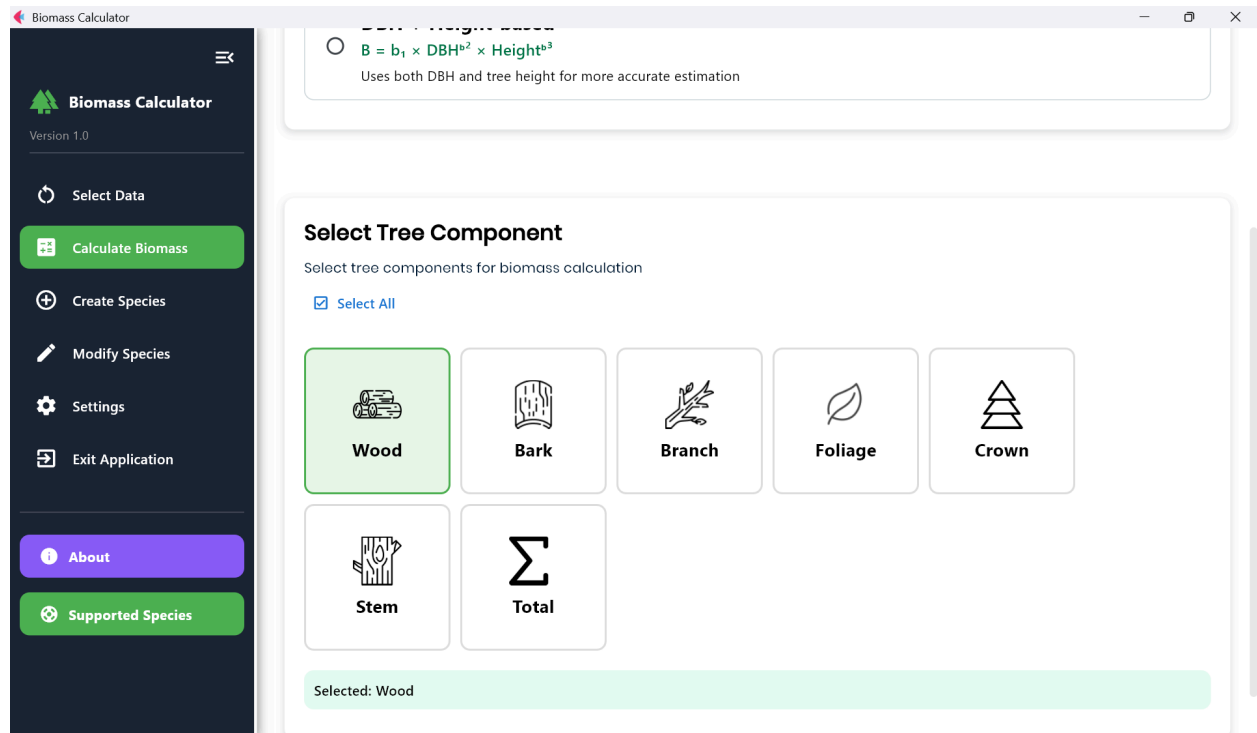


Figure 21. Only “Wood” feature being selected

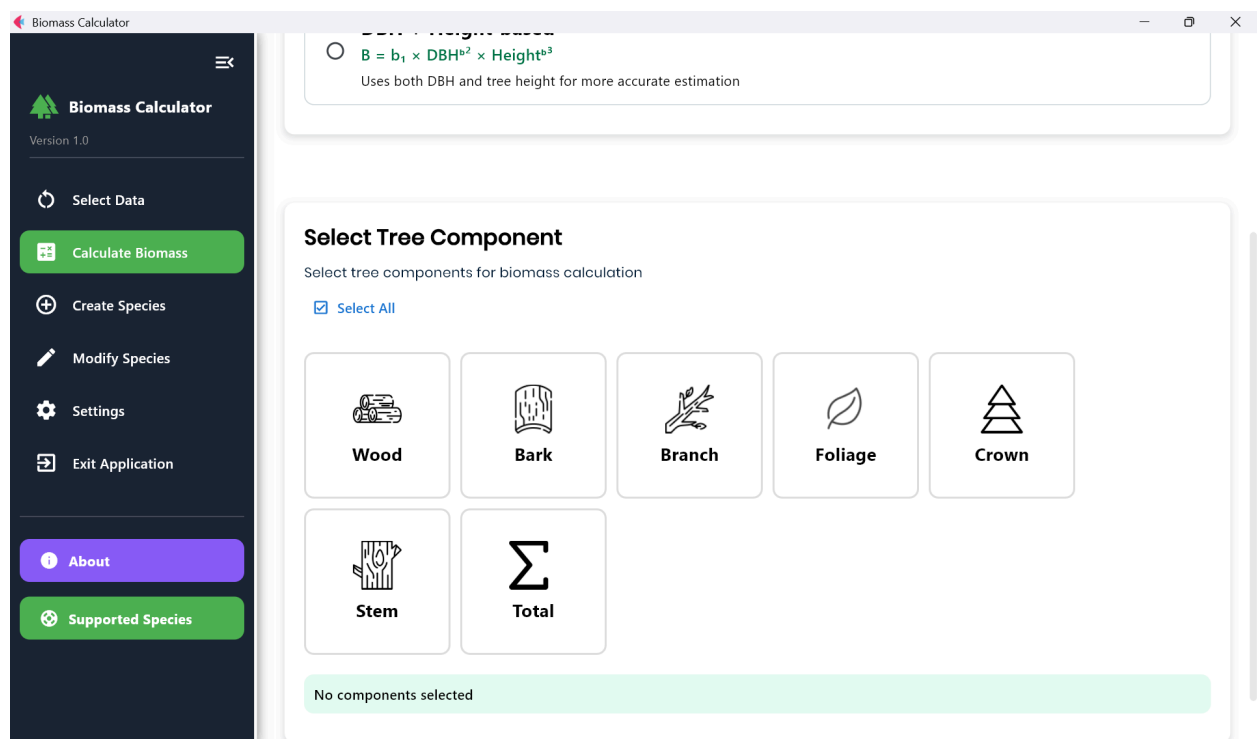


Figure 22. Clicking on “Wood” feature again to deselect it (No feature is selected)

To quickly select all features, click on “Select all” check button. Once all features are being selected, the “Select all” check button will change to “Deselect all” check button that allows users to remove all features from selection.

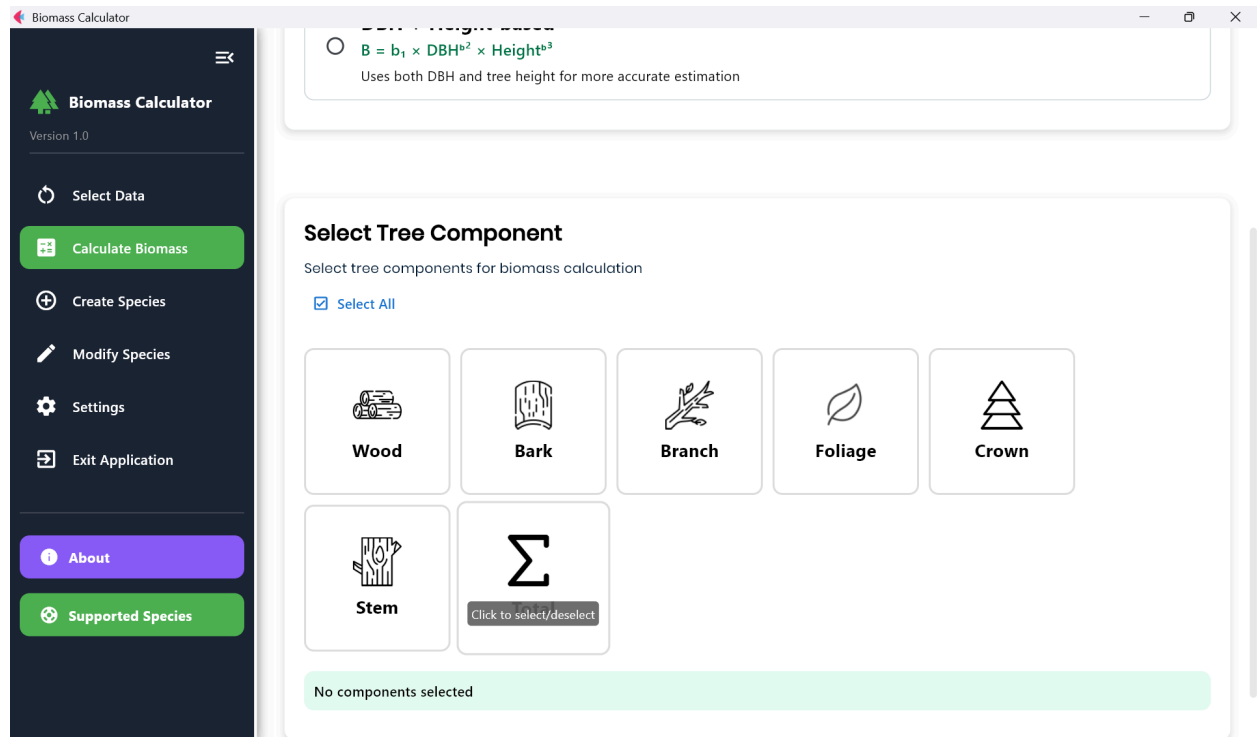


Figure 23. Before clicking “Select All” button (No feature is selected)

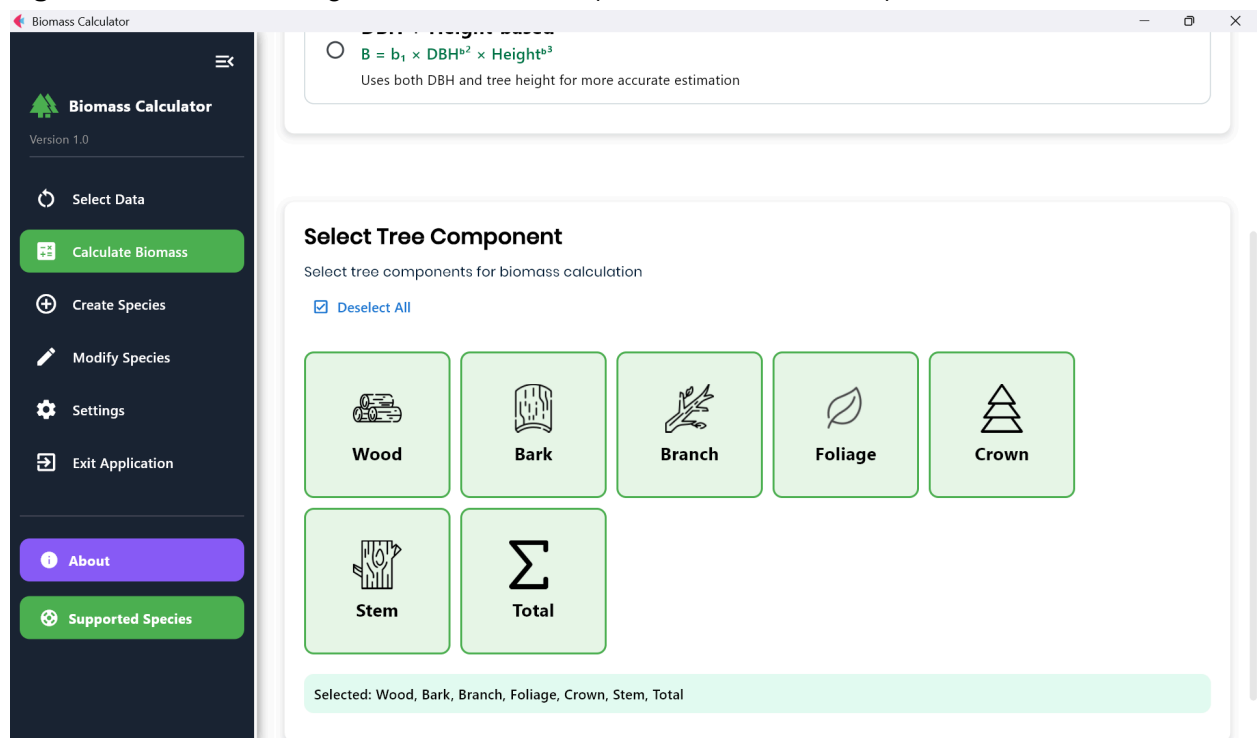


Figure 24. After clicking “Select All” button, the button will change to “Deselect all”

****Note**:** If users import data from a database for calculation, all features will automatically be selected and cannot be chosen to be omitted. In this mode, users can only select DBH-based equation or DBH-and-Height-based equation.

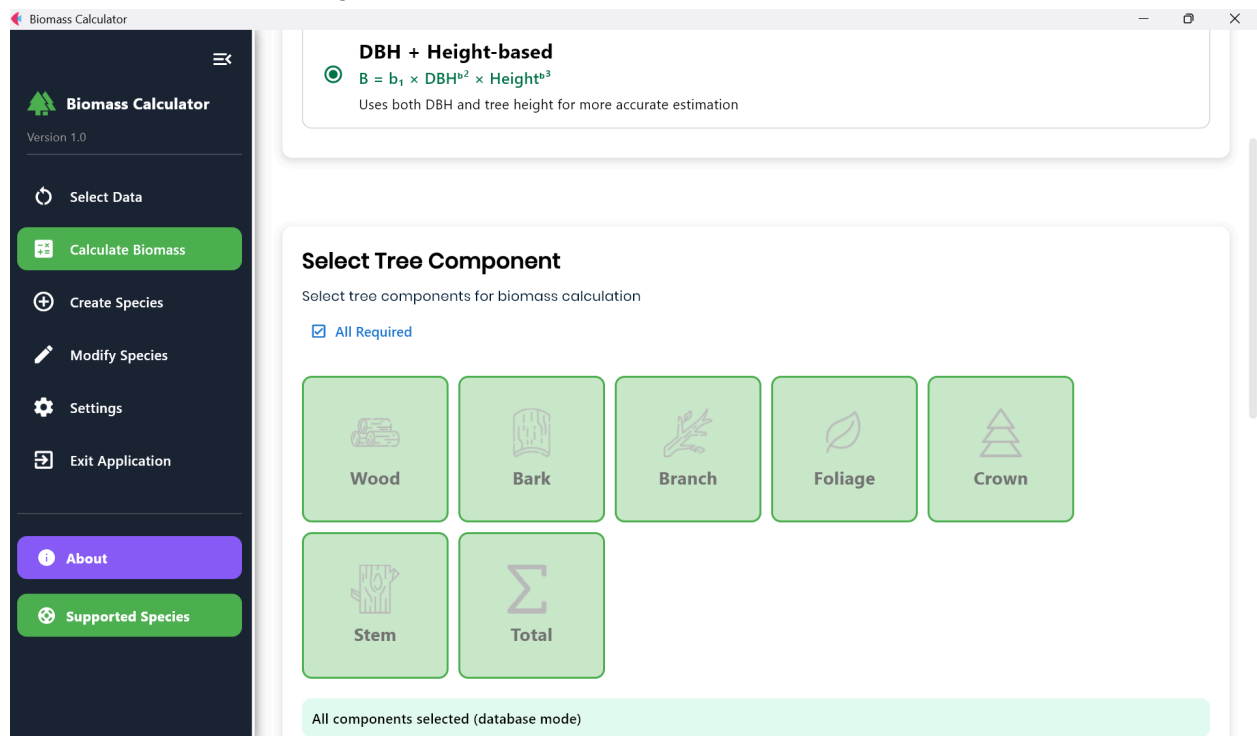


Figure 25. All features must be selected (automatically by default) in database mode

b) Species Classification (Only in Text-file Mode)

After selecting type of function and features for calculation, users can click on the “Calculate Biomass” button at the bottom of the screen. At this point, a pop-up will show up asking users to define the codes for Hardwood and Softwood

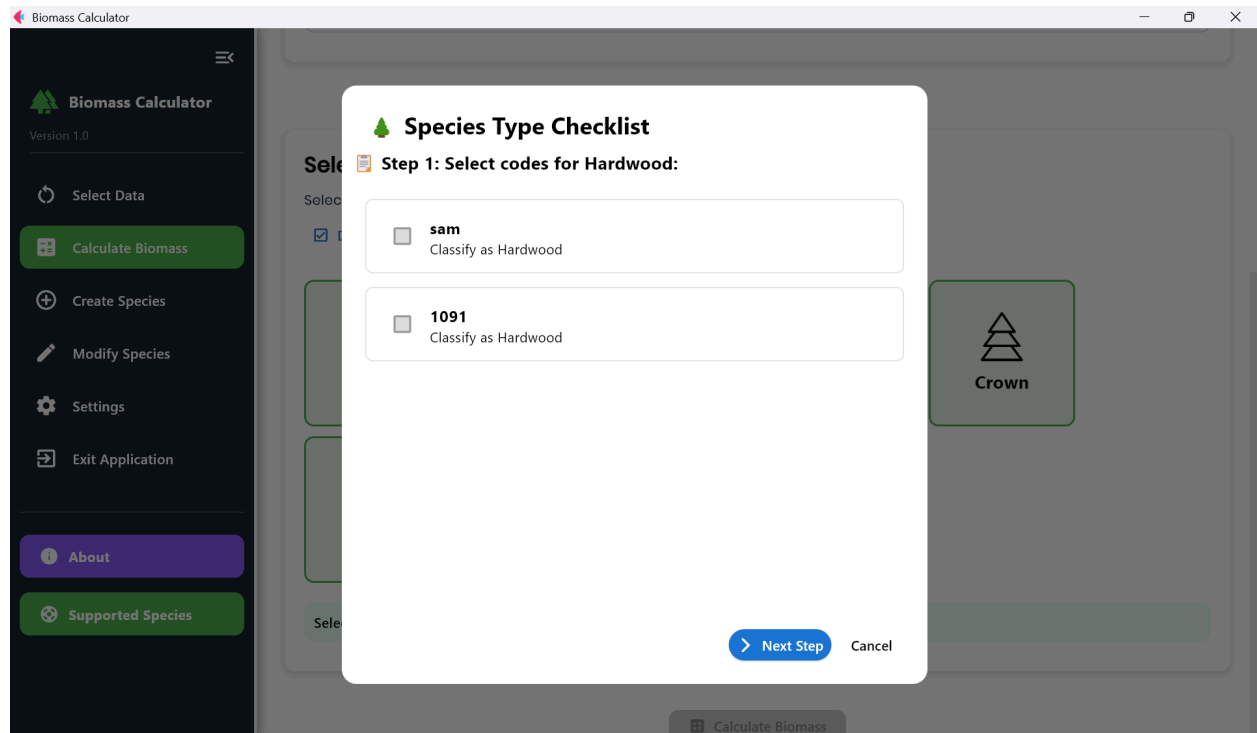


Figure 26. Pop-up after clicking “Calculate Biomass”

Users will be asked to choose one of the two (or more) options as the codename of Hardwood, and can proceed when clicking on “Next Step”. The leftover options after this step will be the available option(s) as codename for Softwood. Choosing the codename for Softwood by clicking on one, and then users can click on “Submit Final” button to start the calculation.

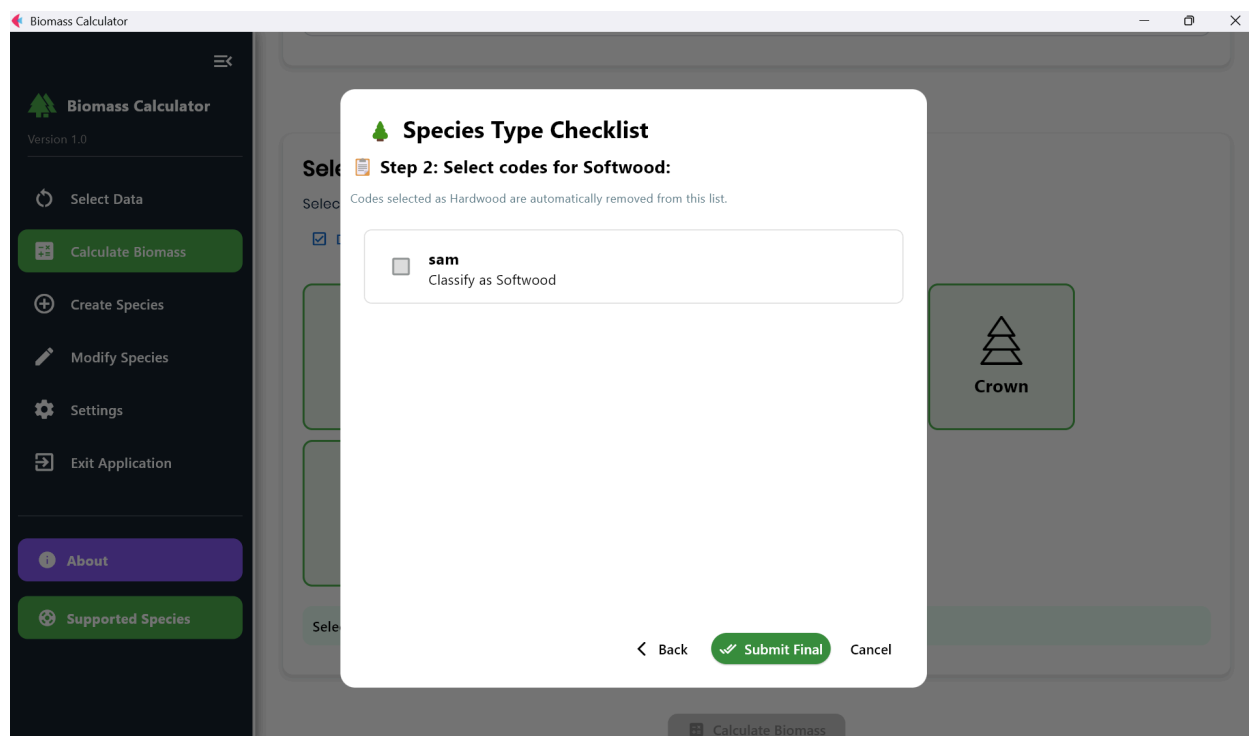


Figure 27. The leftover option(s) after selecting codename for Hardwood. Clicking on “Submit Final” button after choosing a codename for Softwood to start calculating.

c) Exporting Results

After the calculation is done, users can preview the first 10 records of the returned results. Horizontal scroll on the result preview panel is enabled to allow users to see all columns of the generated dataset(s).

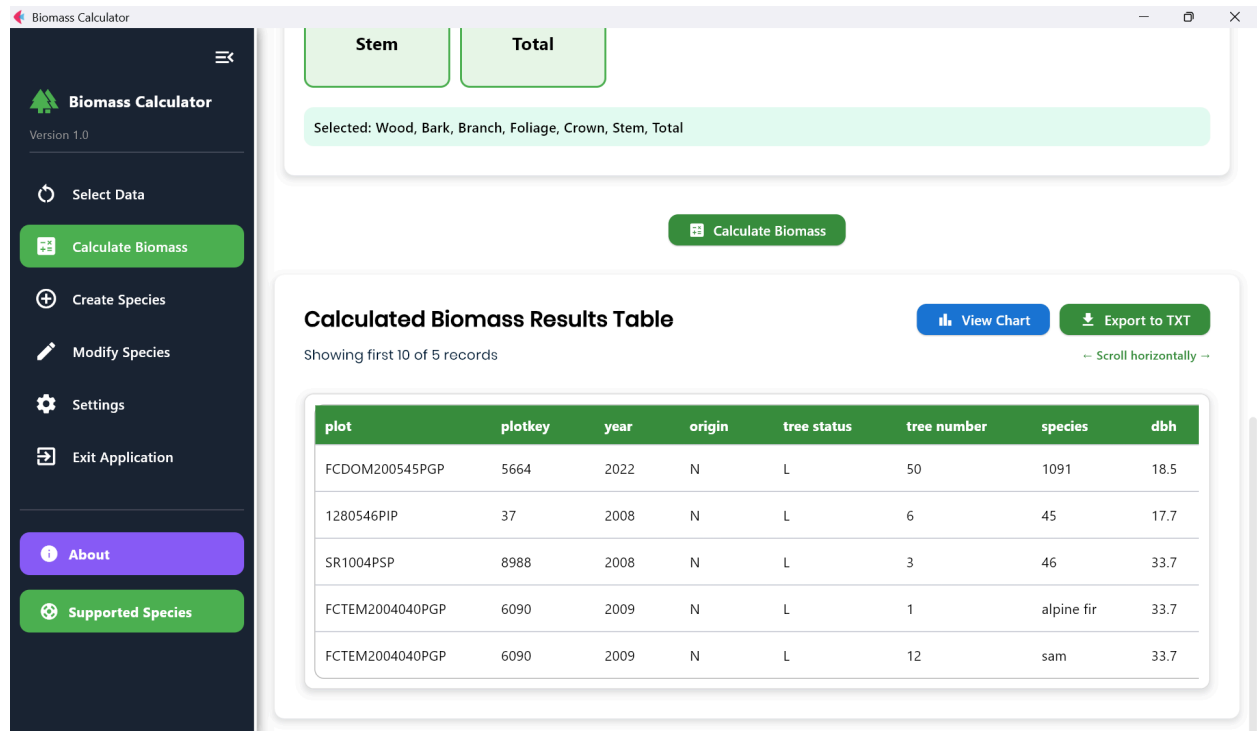


Figure 28. The result preview panel showing the first 10 entries of the generated dataset. Users can scroll horizontally to see all the columns

i) Exporting to Text File

If the input data is from a text file, then users can only export the output data to a new text file. To write the calculated data to a new text file, click on the “Export to TXT” button to open up a file-saving dialogue.

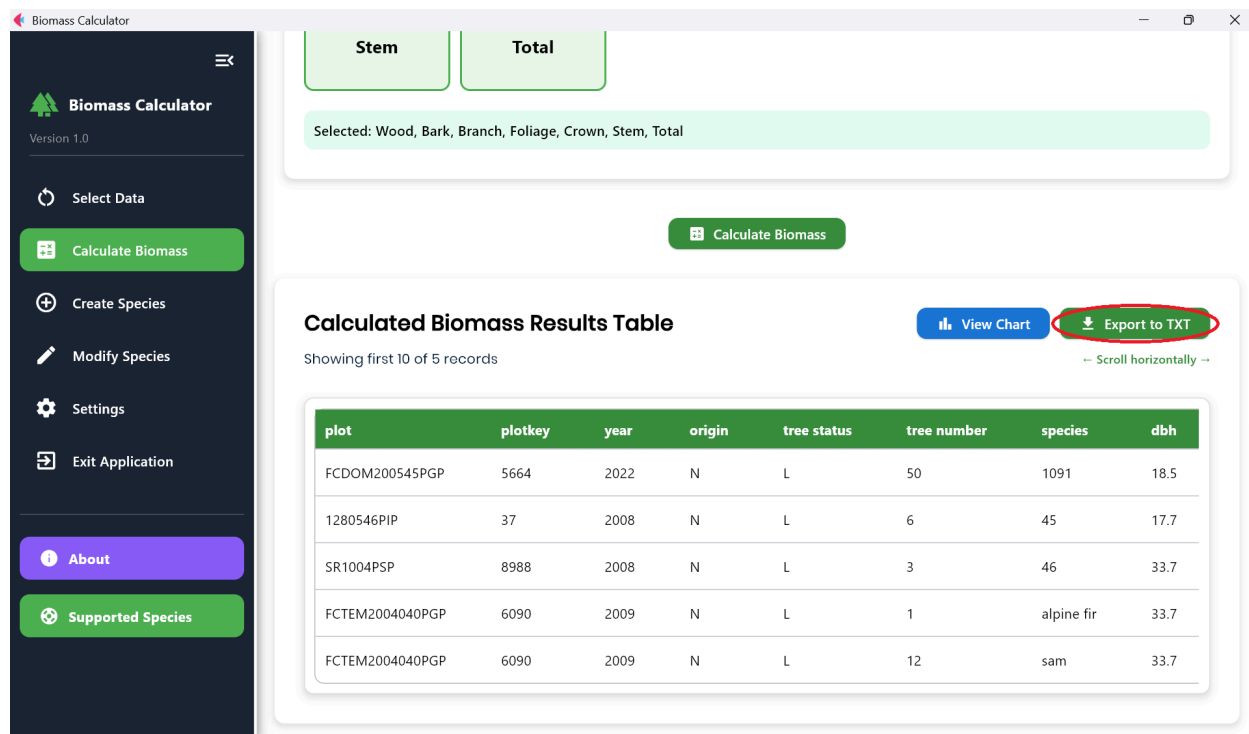


Figure 29. Click on [Export to TXT] button to open file-saving dialogue

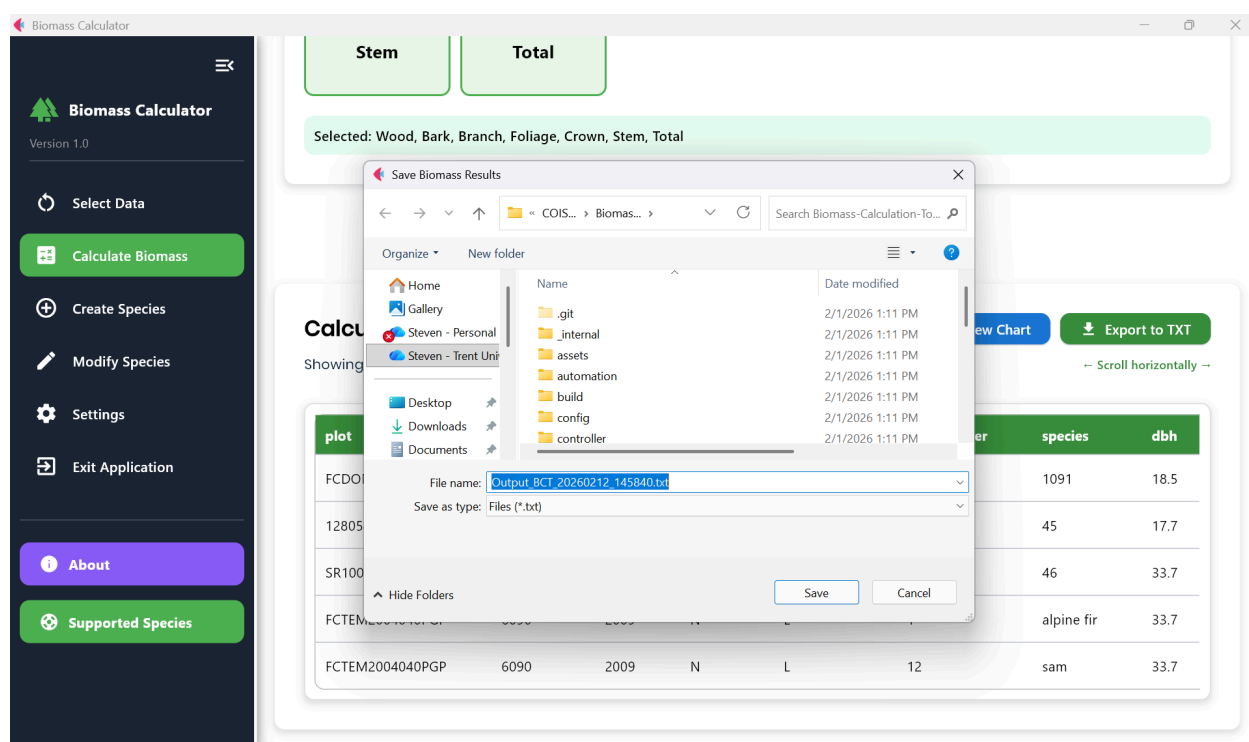


Figure 30. File name is automatically suggested based on input file's name

ii) Writing to Database

This option is only available if the input data came from a database. Clicking on the orange [Write to Database] button will save output data to the same database of input data

The screenshot shows the Biomass Calculator application interface. On the left is a dark sidebar with navigation options: Select Data, Calculate Biomass (highlighted in green), Create Species, Modify Species, Settings, Exit Application, About (purple), and Supported Species (green). The main panel displays the 'Calculated Biomass Results Table' with a 'Calculate Biomass' button at the top. Below the title are three buttons: 'View Chart' (blue), 'Export to TXT' (green), and 'Write to Database' (orange, highlighted with a red circle). The table shows the first 10 of 10000 records. The table has columns: plot, year, species, tree_number, dbh, height, Wood (KG), and Bark (KG). The data rows show various tree measurements for plot 1246201PIP in 2007.

plot	year	species	tree_number	dbh	height	Wood (KG)	Bark (KG)
1246201PIP	2007	38	39	11.3	15.4	33.5381	5.9444
1246201PIP	2007	38	40	13.5	16.5	50.8248	8.8445
1246201PIP	2007	38	44	14.8	17.0	62.7551	10.8188
1246201PIP	2007	12	45	11.7	10.2	21.0147	3.6288
1246201PIP	2007	13	47	5.1	5.1	2.7459	0.5734
1246201PIP	2007	20	49	2.9	3.2	0.5670	0.0965
1246201PIP	2007	38	50	16.8	17.7	83.8800	14.2755
1246201PIP	2007	20	51	4.0	4.1	1.2676	0.2183
1246201PIP	2007	20	52	3.2	3.4	0.7159	0.1228
1246201PIP	2007	38	53	9.5	14.2	22.1469	3.9976

Figure 31. Preview panel after the biomass calculation is done. Choose [Write to Database] to save data to preconfigured table

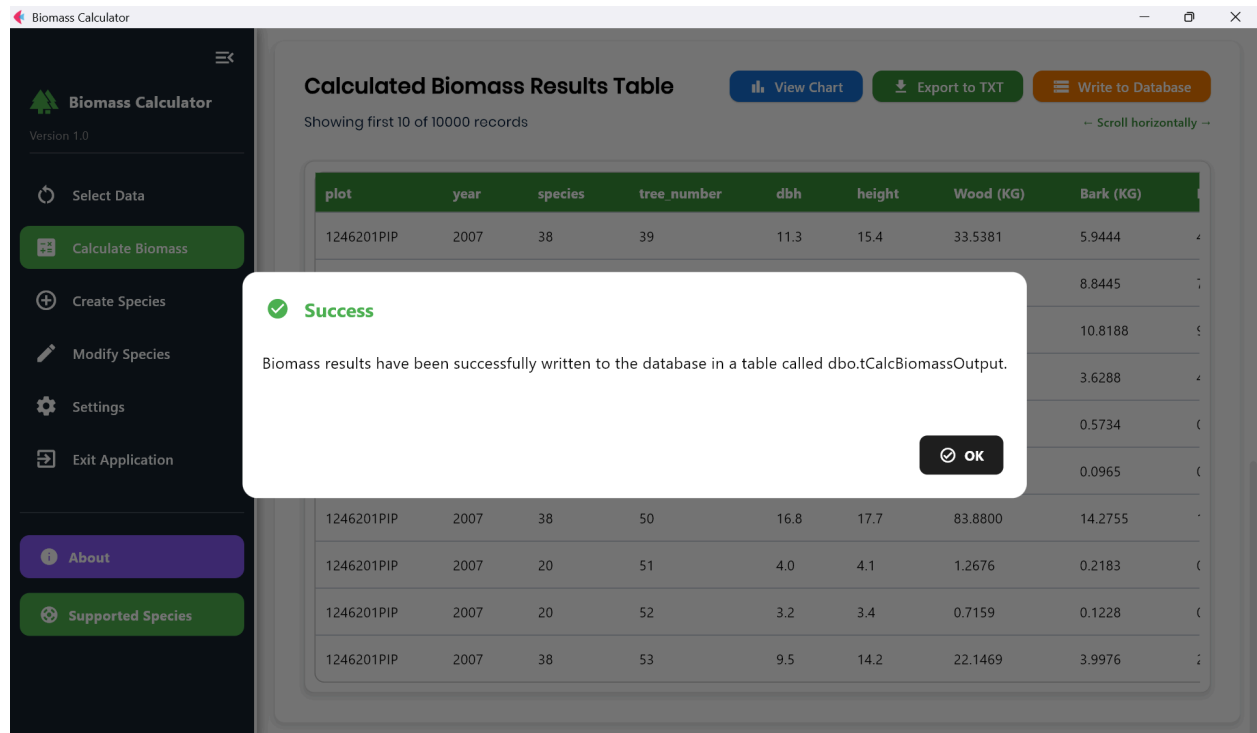


Figure 32. Notice when output data is successfully recorded in database

iii) Generating Plot

Users can see plot(s) mapped from data points in the calculated output. To view the graph(s), click on the blue [View Chart] button, and the graph(s) can be saved as individual files.

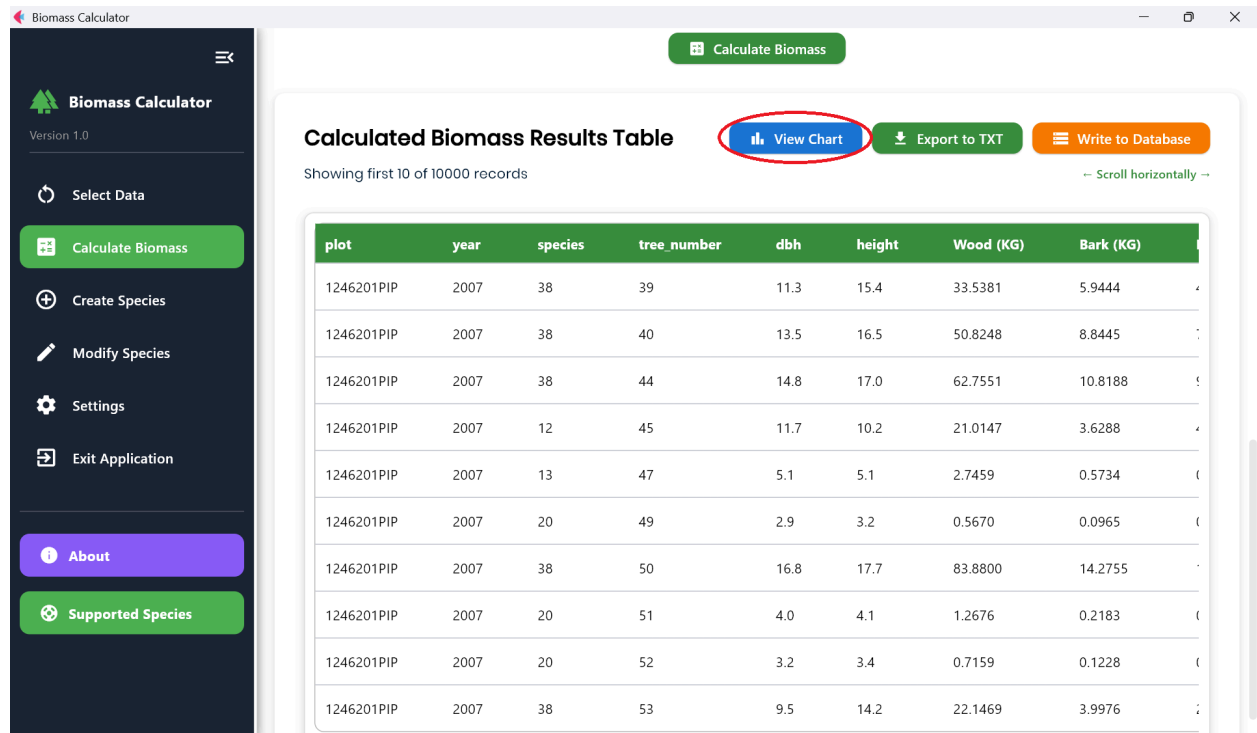


Figure 33. Click [View Chart] button to see generated graph(s) from data points from output

The Download icon on the top right corner of the chart panel allows users to save the graph(s) as file(s). To quit that panel, click on the [X] button to the right of Download icon to go back to the output preview panel.

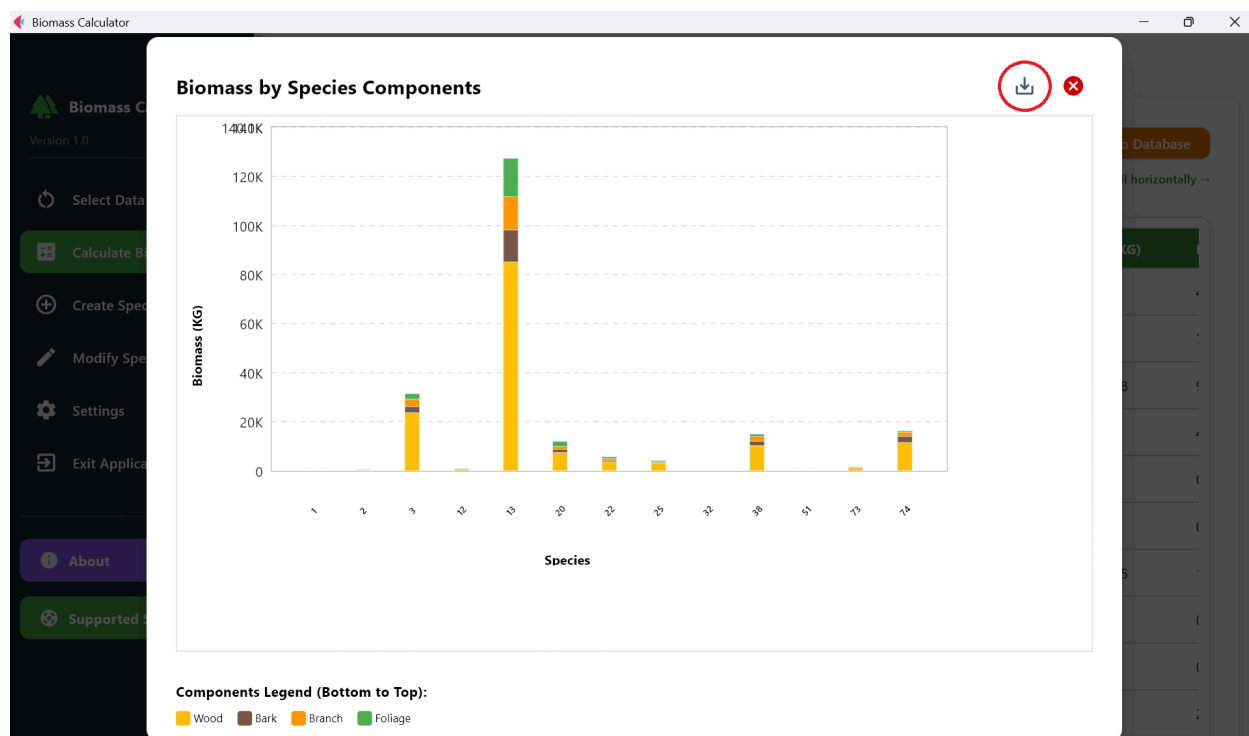


Figure 34. Click Download icon (red circle highlight) to save chart as an individual file

d) Missing Species?

If the input data has entries containing tree species designation (name, code number) unrecognized by the tool, the output data can become full of errors or skewed values. When this happens, users can [create new Species](#) entries for the tool's memory to ensure biomass calculations are accurate

5) Create Species

To go to the "Create Species" console, click on the [Create Species] button on the left-hand menu and users will be shown the "Create Species" screen

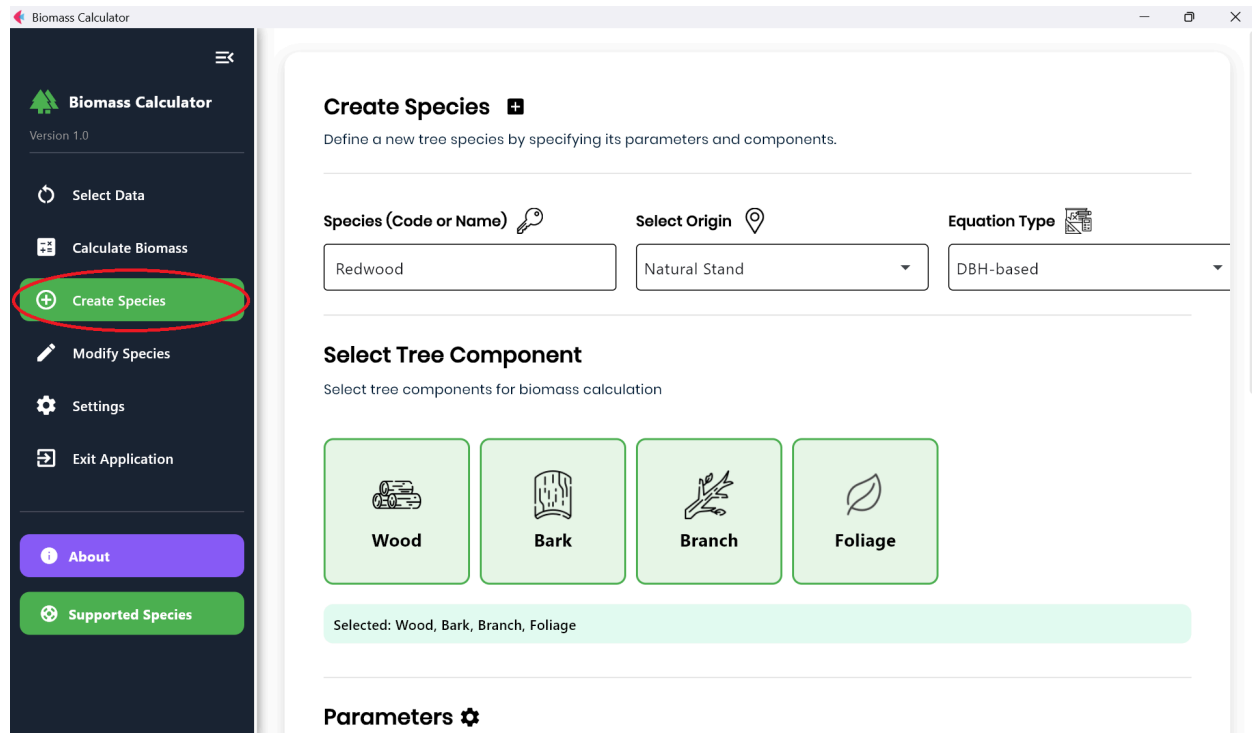


Figure 35. Click [Create Species] button (red outline) to show the console

Users will be asked to put in mandatory details of the new tree species (Name or Code number, Origin, Applicable Equation type, tree features to be accounted for in biomass function, and parameter values corresponding to select features for calculation)

In the “Parameters” sections, all values to fill must be in the (-5.0, 5.0) range inclusively. If users omit or enter invalid values, warning messages in red will appear at the bottom of the respective fields.

The screenshot shows the 'Biomass Calculator' application interface. On the left is a dark sidebar with navigation options: 'Select Data', 'Calculate Biomass', 'Create Species' (highlighted in green), 'Modify Species', 'Settings', 'Exit Application', 'About' (highlighted in purple), and 'Supported Species' (highlighted in green). The main content area has a light gray background. At the top, there are three input fields: 'Species (Code or Name)' with the value 'Alpine fir or 123' and a red warning message 'It must be a number or a name (1-30 character...)', 'Select Origin' with a dropdown menu showing 'Natural Stand', and 'Equation Type' with a dropdown menu showing 'DBH-based'. Below these is a section titled 'Select Tree Component' with the instruction 'Select tree components for biomass calculation'. It contains four green buttons with icons: 'Wood' (logs), 'Bark' (bark cross-section), 'Branch' (branch with leaves), and 'Foliage' (leaf). Below the buttons is a green bar indicating 'Selected: Wood, Bark, Branch, Foliage'. At the bottom is a 'Parameters' section with a gear icon. It contains two input fields: 'b1' with the value '5.90' and a red warning message 'Value must be between -5.0 and 5.0', and 'b2' with the value '459.0' and a red warning message 'Value must be between -5.0 and 5.0'.

Figure 36. Warning messages appear when fields left empty or filled with invalid values

If all necessary fields are filled with acceptable values, a preview panel will pop up to allow users to see all details of the newly-entered species to make sure everything is correct. Click on the [Proceed] button at the bottom-right corner to save the new tree species, or [Cancel] to go back to the “Create Species” console

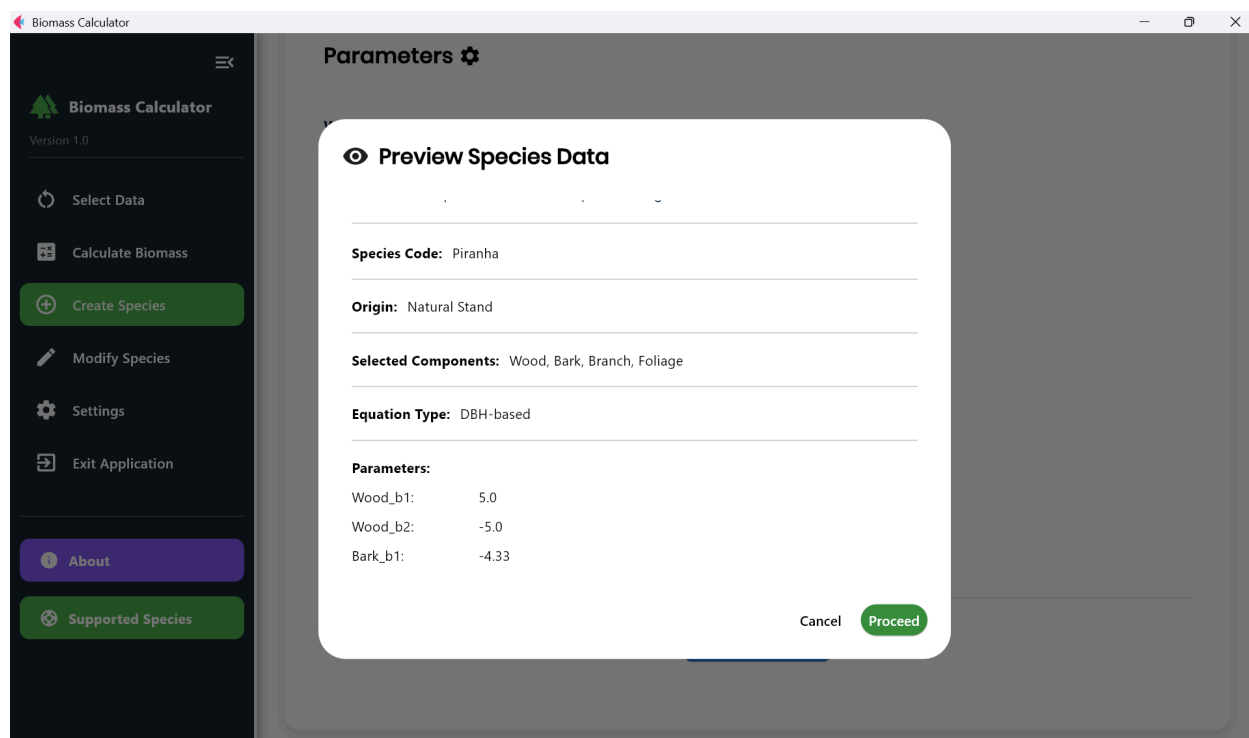


Figure 37. Preview panel before saving a new tree species.

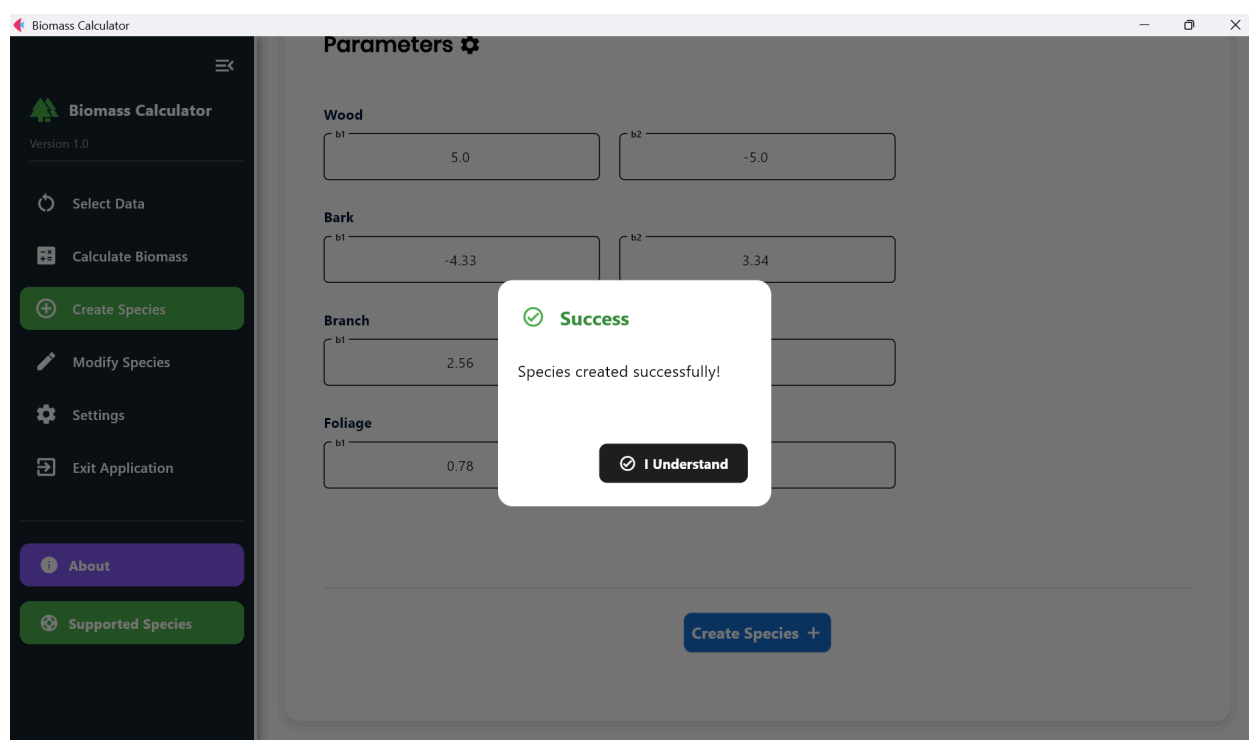


Figure 38. Notice when new tree species added to the tool's memory successfully

6) Modify Species

To modify a created species, open the “Modify Species” menu by clicking on the Modify Species button on the left sidebar.

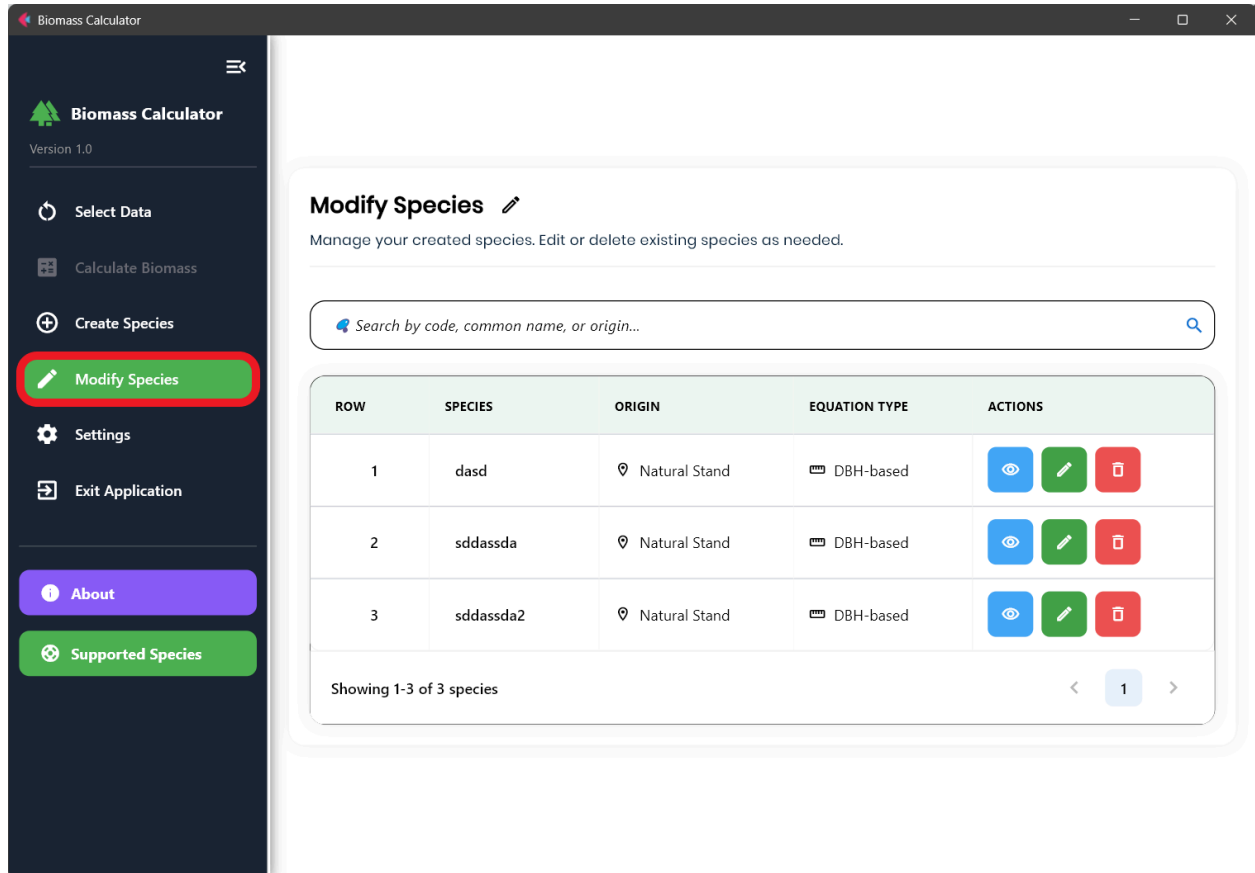


Figure 39. The [Modify Species] button is highlighted.

a) Search for Species

To search for a specific species enter the code, common name, or origin of the species into the search bar.

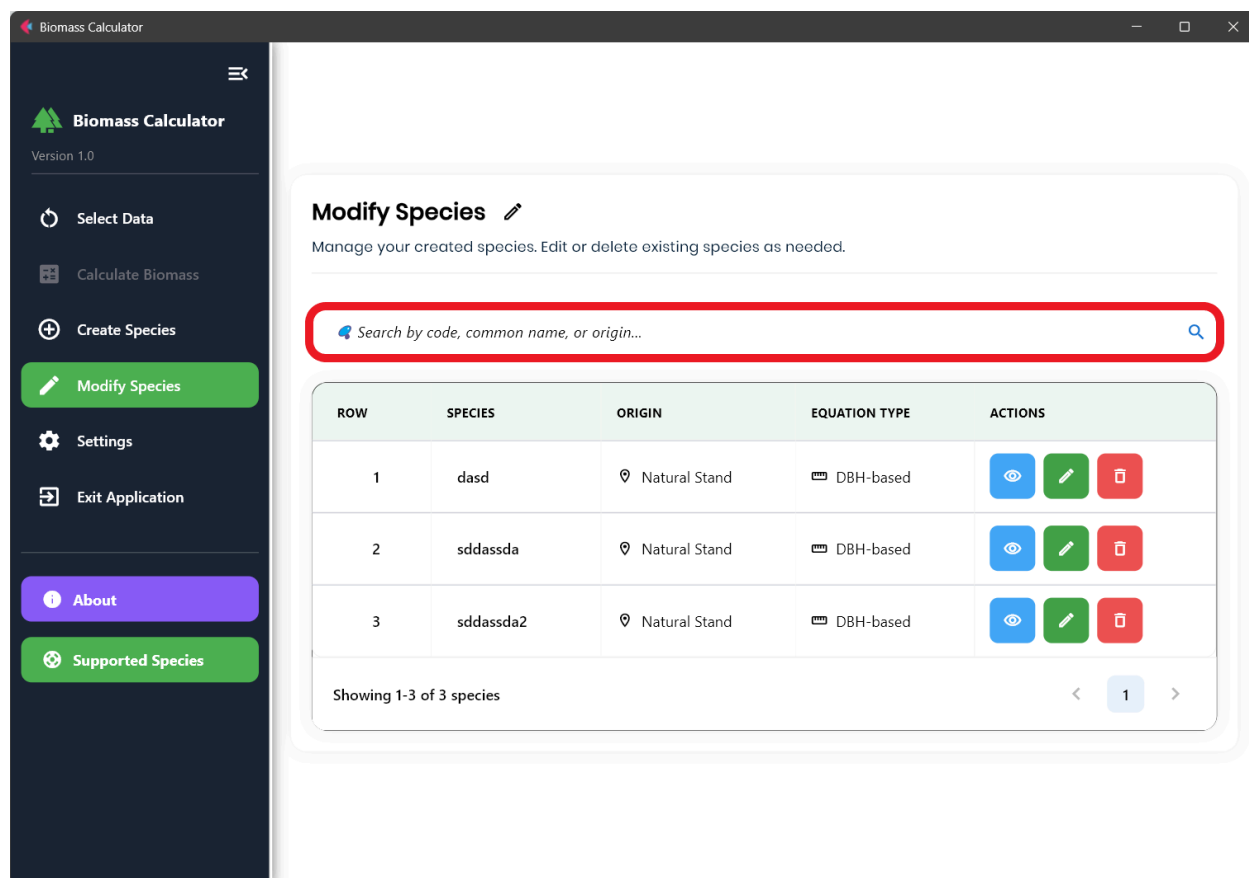


Figure 40. The Modify Species search bar is highlighted.

b) Species Details

To view a specific species' details click on the blue Species Details button indicated by an eyeball.

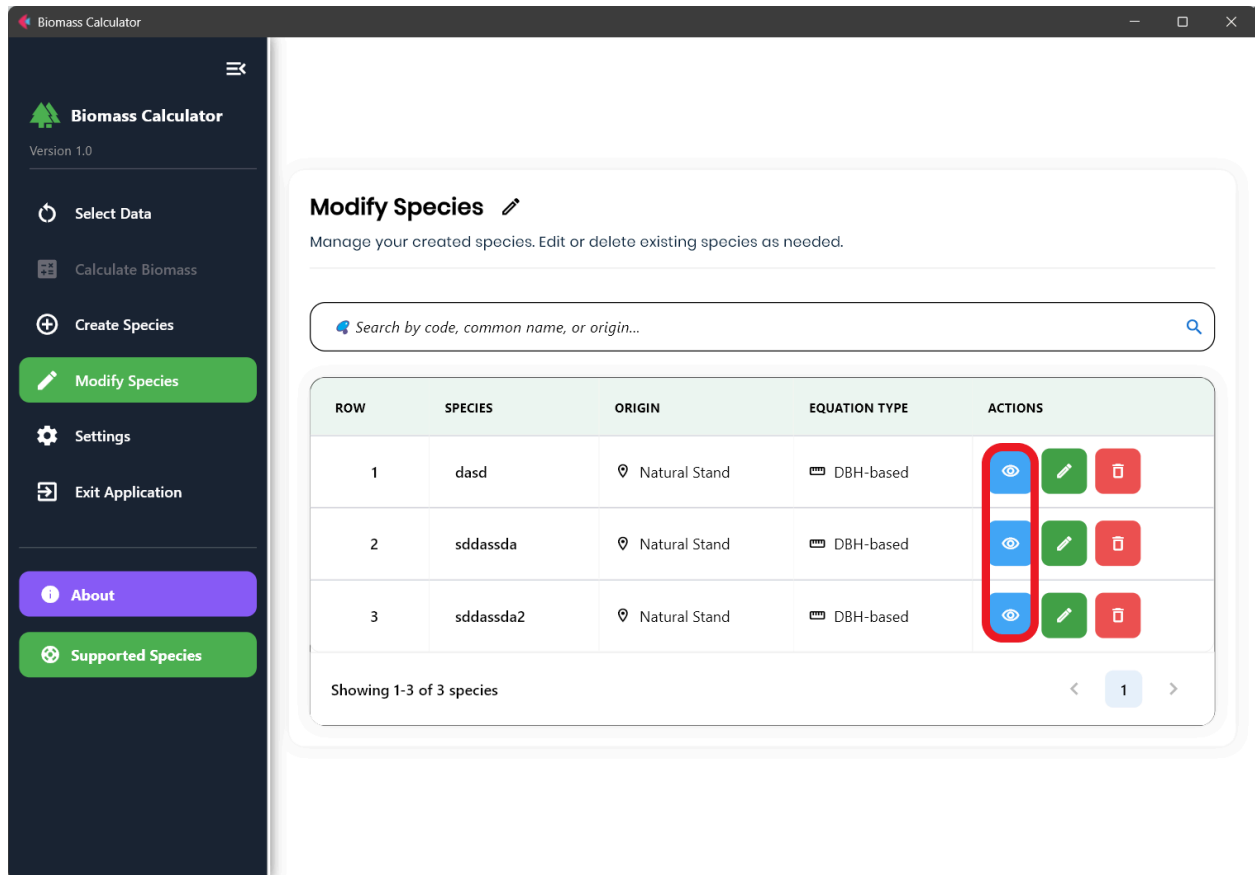


Figure 41. The [Species Details] buttons are highlighted.

You can scroll through the menu that pops up to view all the details for that species.

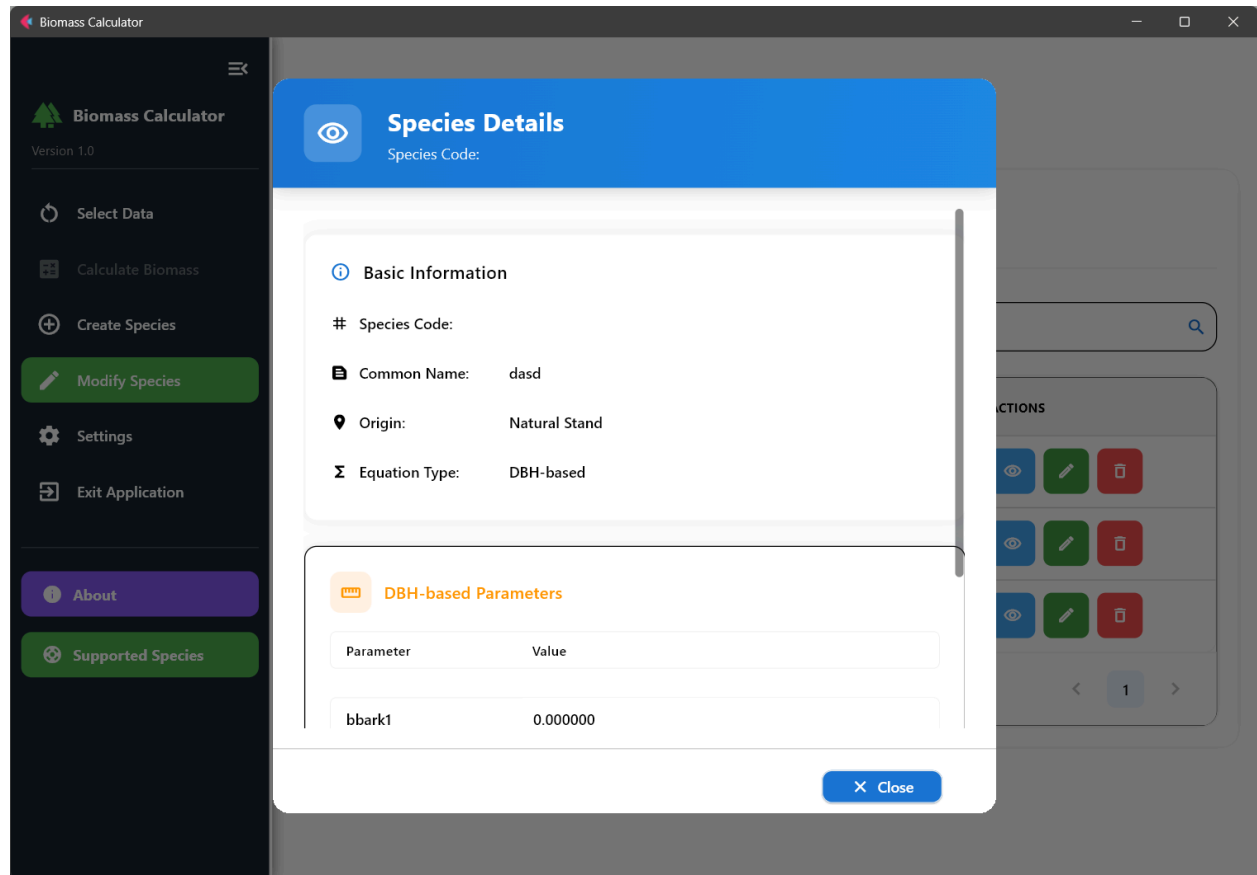


Figure 42. The Species Details menu.

c) Edit Species

To edit a specific species click on the green Edit Species button indicated by a pencil.

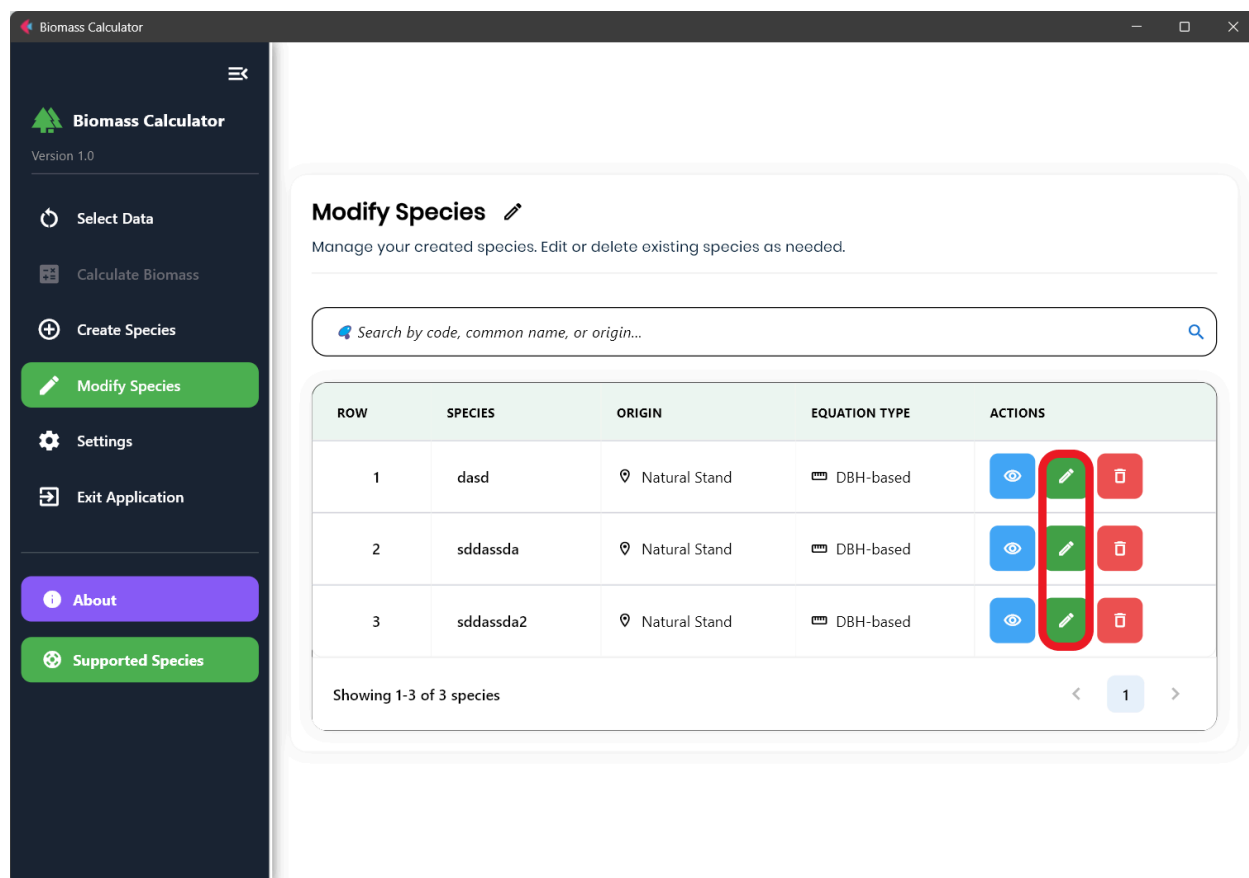


Figure 43. The Edit Species buttons are highlighted.

You can scroll through the menu that pops up and edit any of the details of that species.

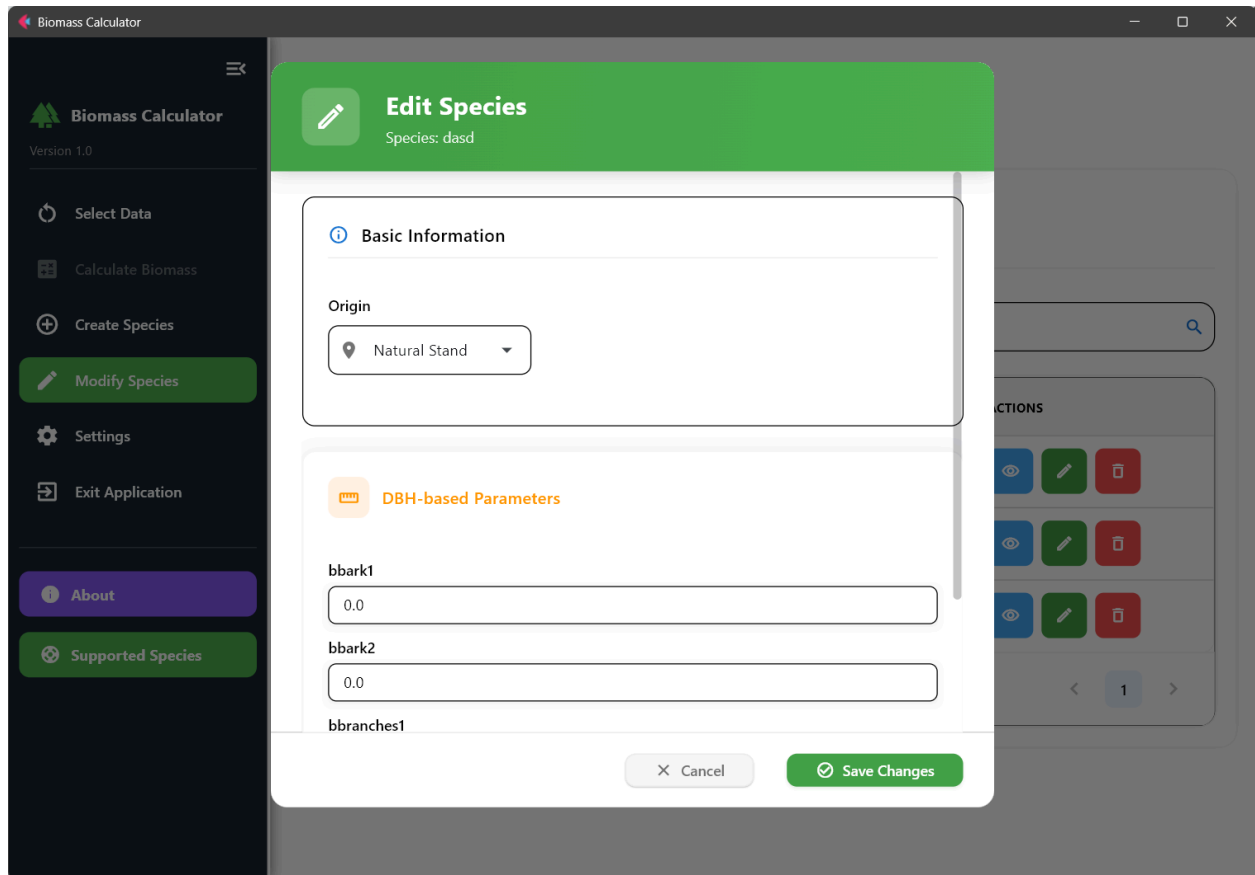


Figure 44. The Edit Species menu.

d) Delete Species

To delete a specific species click on the red Delete Species button indicated by a garbage can.

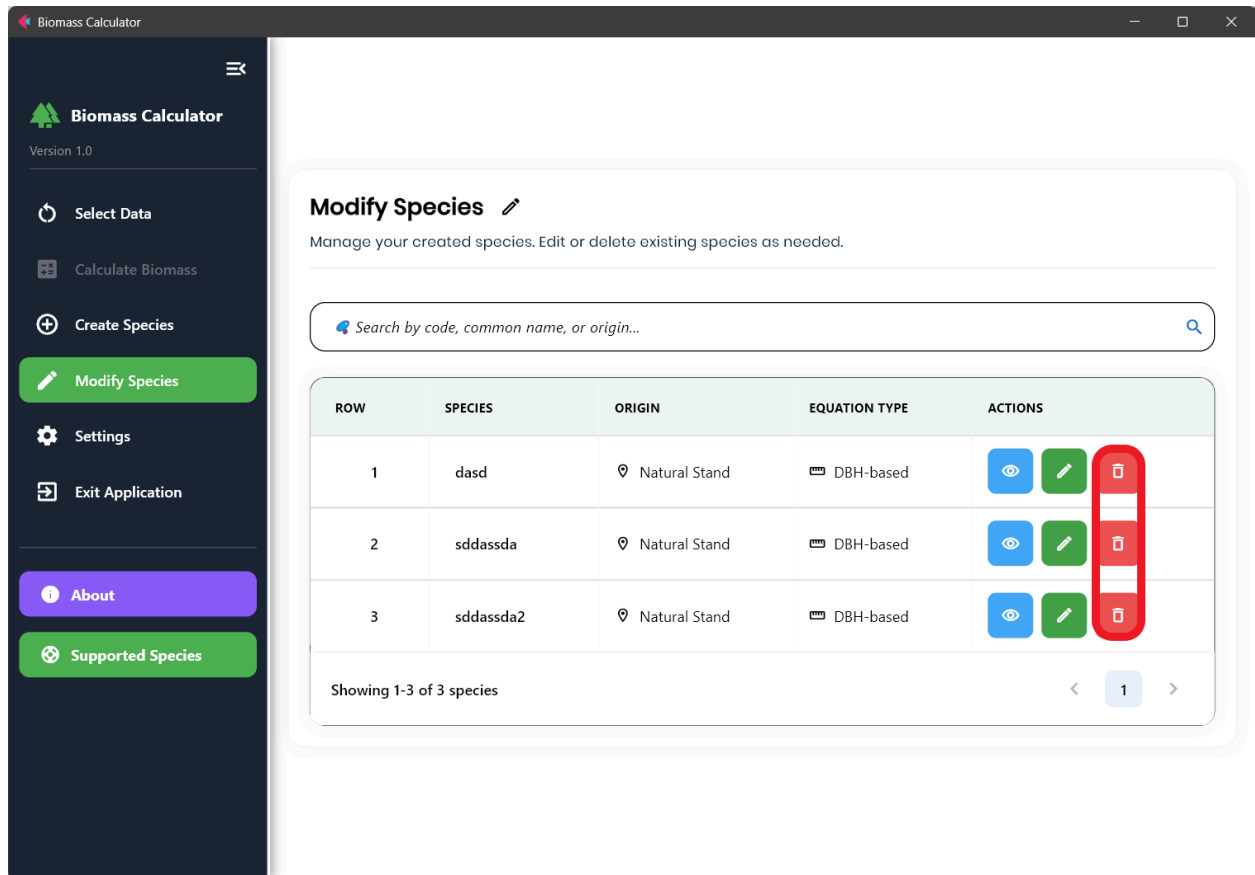


Figure 45. The Delete Species buttons are highlighted.

You then have to confirm the delete by pressing on the “Delete Species” button on the bottom of the menu.

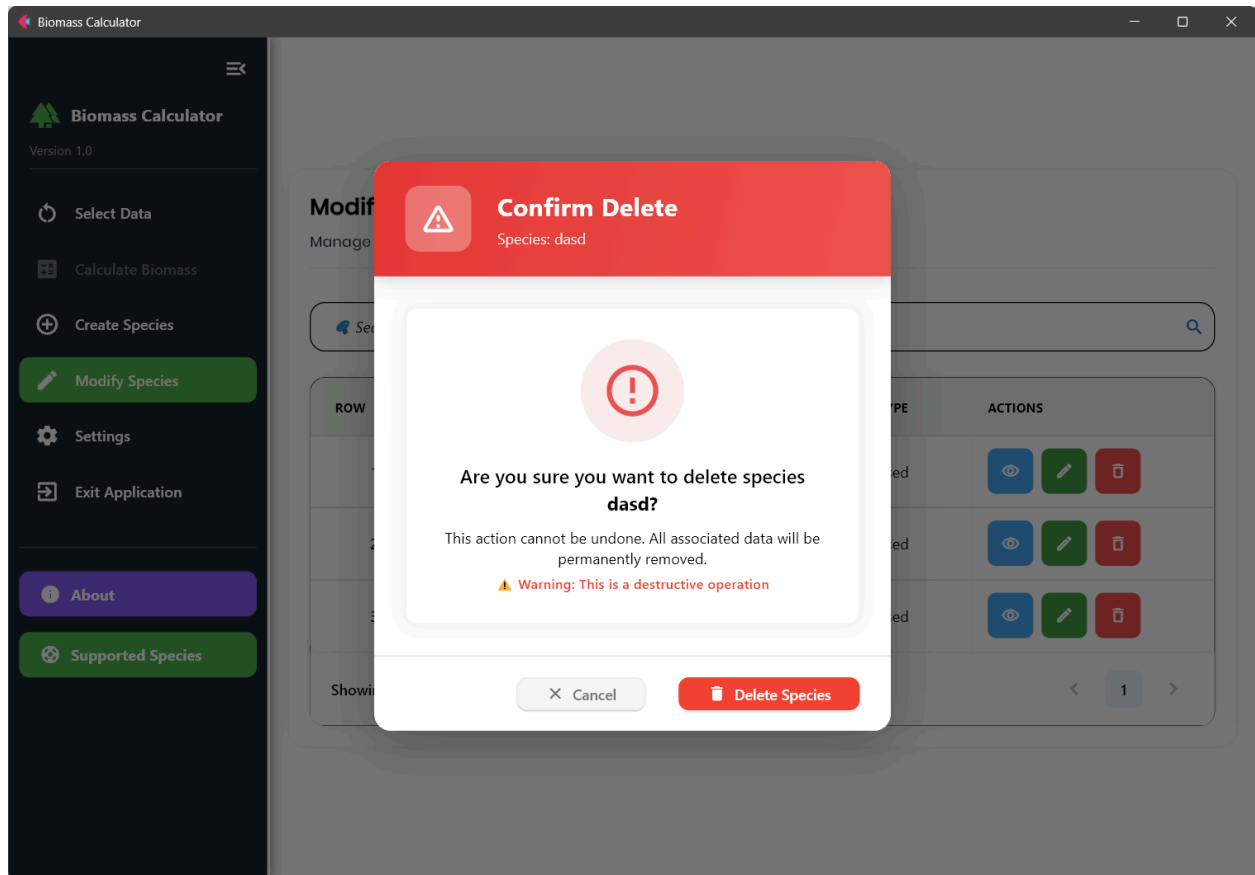


Figure 46. The Confirm Delete menu.

7) Settings

To change any settings, open the “Settings” menu by clicking on the Settings button on the left sidebar.

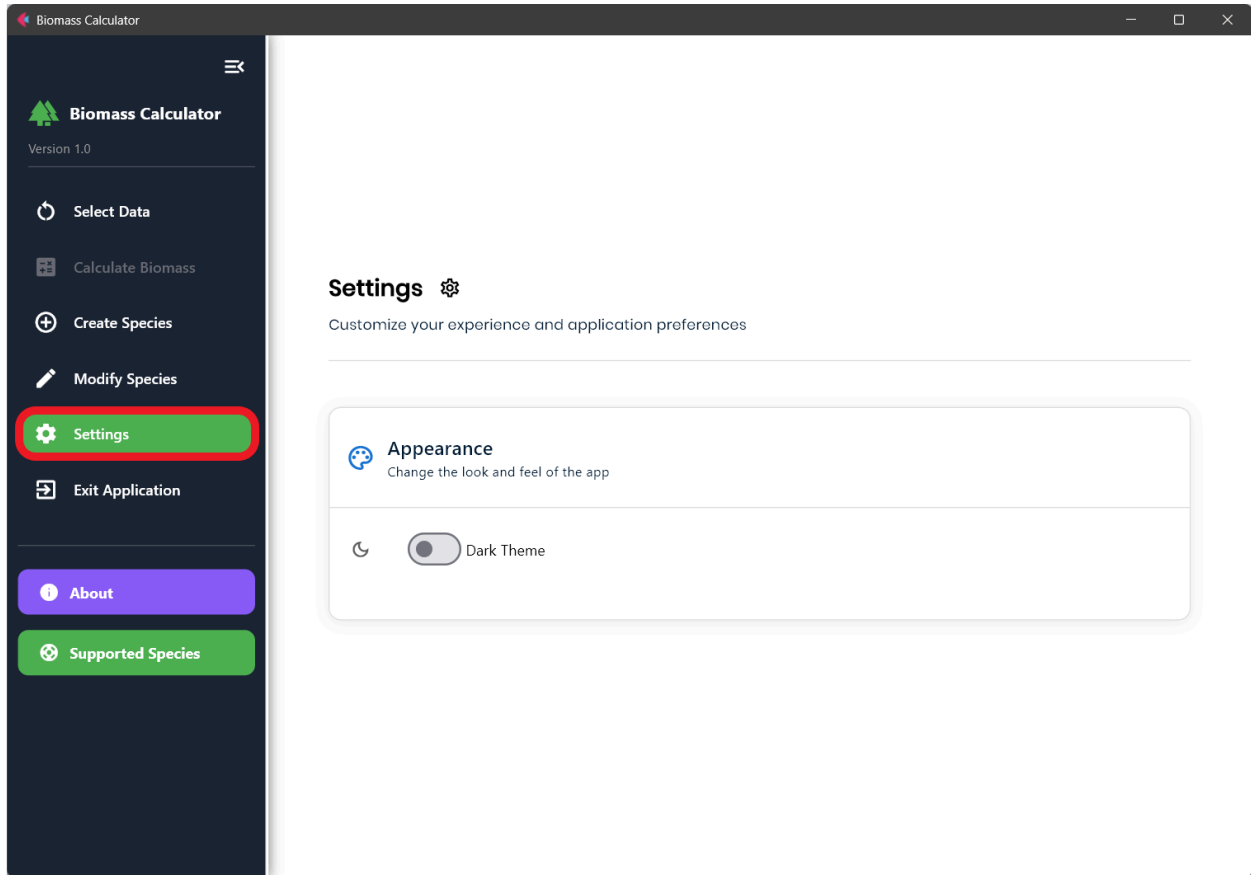


Figure 47. The [Settings] button is highlighted.

The only setting currently is to change the theme from Light to Dark.

8) Exit Application

To exit the application just click on the “Exit Application” button on the left sidebar.

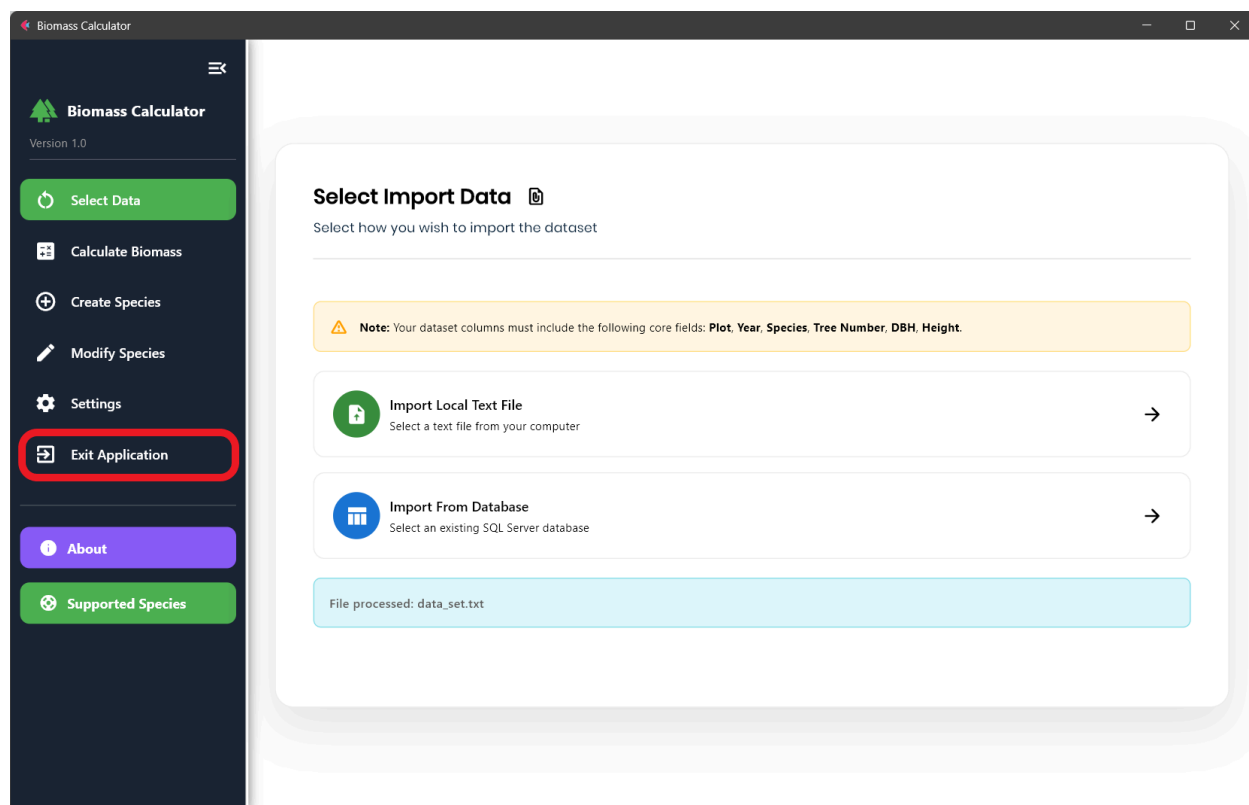


Figure 48. The [Exit Application] button is highlighted.

Then click on the “Exit Anyway” button on the menu that shows up.

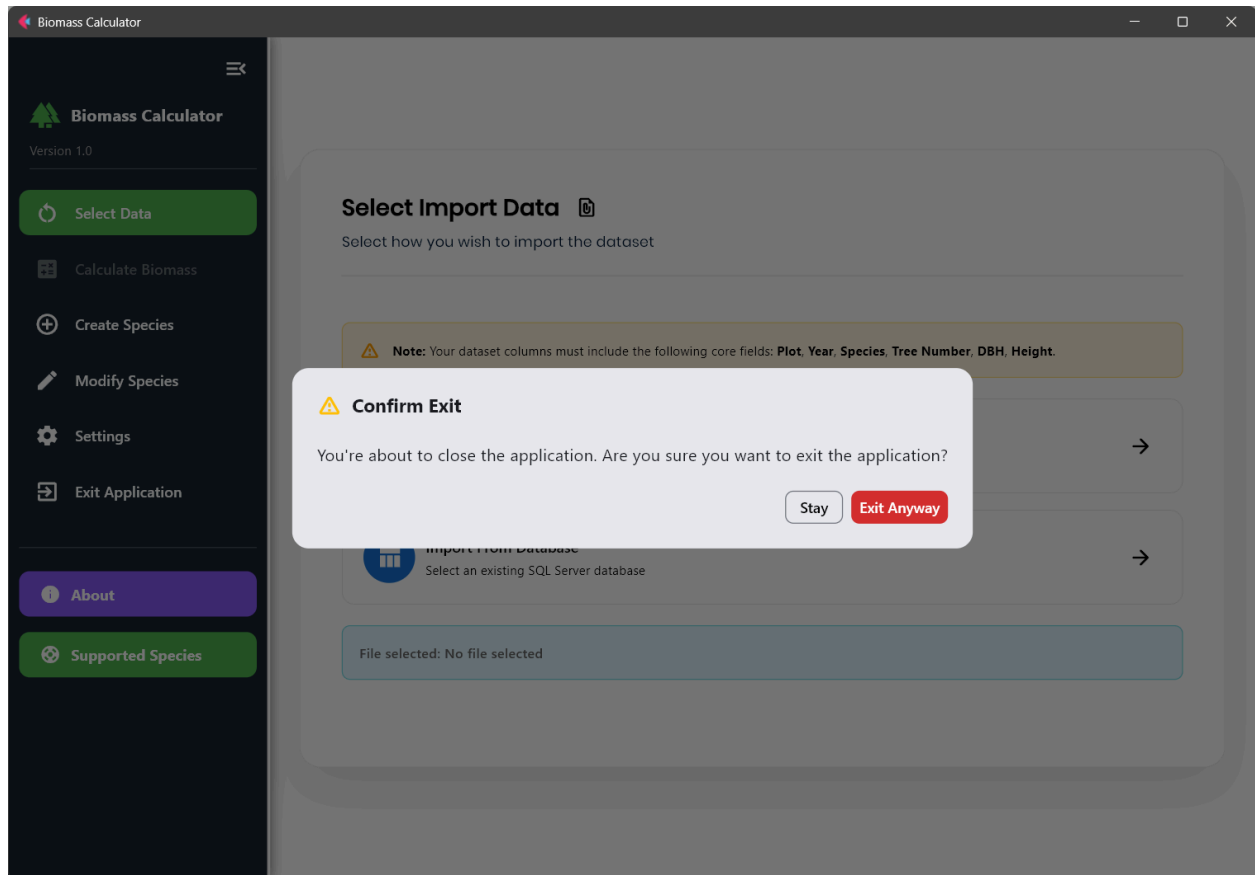


Figure 49. The Confirm Exit menu.

Alternatively, just click on the window's X in the top right.

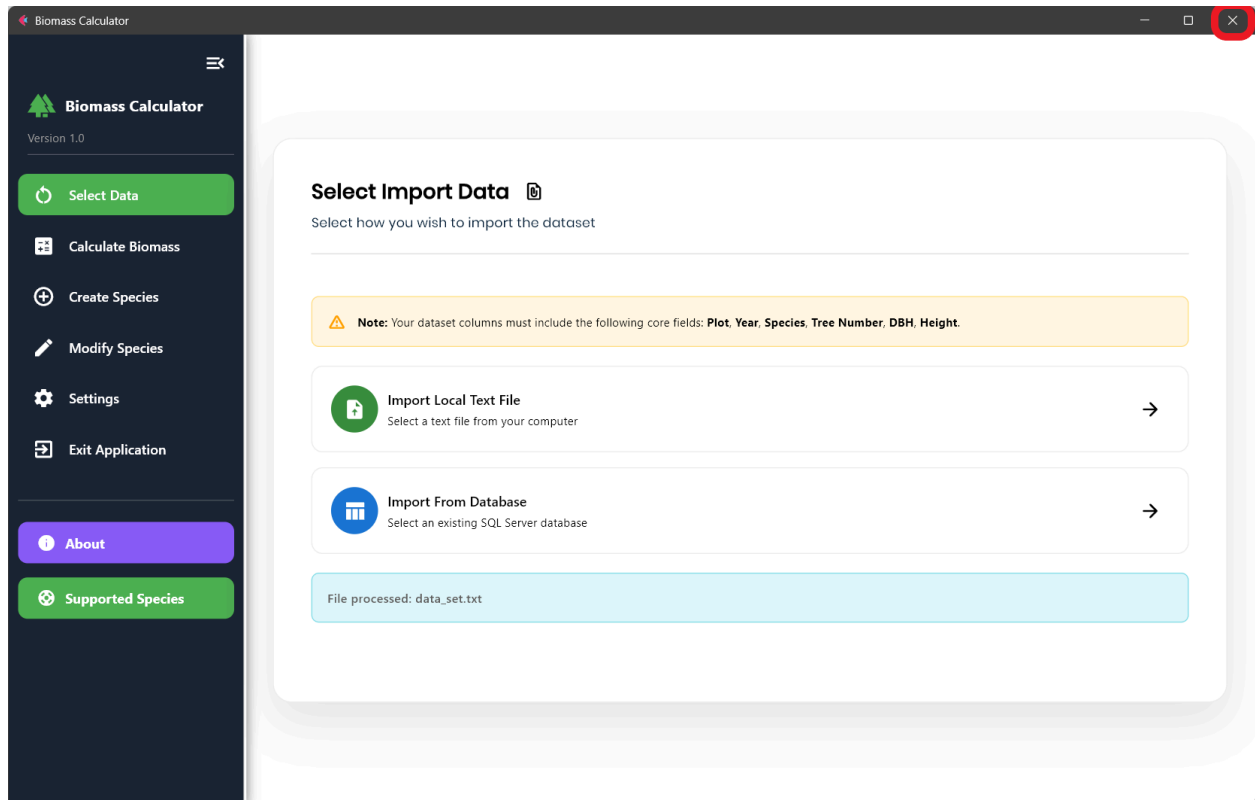


Figure 50. The window X button is highlighted.

9) Log

When you are done with your session and exit the calculator, you can go to the “log” directory (which is in the same directory as the program) to find dated text files with timestamped records of your activities for that session and all previous sessions.